

## CHAPTER 8: LINEAR TRANSFORMATIONS

### 8.1 General Linear Transformations

1. (a)  $T\left(2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = T\left(\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\right) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}^2 = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$  does not equal  $2T\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$  so  $T$  does not satisfy the homogeneity property. Consequently,  $T$  is not a linear transformation.

- (b) Let  $A$  and  $B$  be any  $2 \times 2$  matrices and let  $k$  be any real number. We have

$$T(kA) = \text{tr}(kA) = ka_{11} + ka_{22} = k(a_{11} + a_{22}) = k \text{tr}(A) = kT(A) \text{ and}$$

$$\begin{aligned} T(A+B) &= \text{tr}(A+B) = (a_{11} + b_{11}) + (a_{22} + b_{22}) = (a_{11} + a_{22}) + (b_{11} + b_{22}) \\ &= \text{tr}(A) + \text{tr}(B) = T(A) + T(B) \end{aligned}$$

therefore  $T$  is a linear transformation.

The kernel of  $T$  consists of all matrices  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  such that  $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a + d = 0$ , i.e.,  $d = -a$ .

We conclude that the kernel of  $T$  consists of all matrices of the form  $\begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ .

- (c) Let  $A$  and  $B$  be any  $2 \times 2$  matrices and let  $k$  be any real number. We have

$$T(kA) = kA + (kA)^T = kA + kA^T = k(A + A^T) = kT(A) \text{ and}$$

$$T(A+B) = A + B + (A+B)^T = A + B + A^T + B^T = A + A^T + B + B^T = T(A) + T(B)$$

therefore  $T$  is a linear transformation.

The kernel of  $T$  consists of all matrices  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  such that

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a & b \\ c & d \end{bmatrix}^T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 2a & b+c \\ b+c & 2d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ therefore}$$

$$a = d = 0 \text{ and } c = -b.$$

We conclude that the kernel of  $T$  consists of all matrices of the form  $\begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix}$ .

2. (a) Let  $A$  and  $B$  be any  $2 \times 2$  matrices and let  $k$  be any real number. We have

$$T(kA) = (kA)_{11} = ka_{11} = k(A)_{11} = kT(A) \text{ and}$$

$$T(A+B) = (A+B)_{11} = a_{11} + b_{11} = (A)_{11} + (B)_{11} = T(A) + T(B)$$

therefore  $T$  is a linear transformation.

The kernel of  $T$  consists of all matrices  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  such that  $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a = 0$ .

We conclude that the kernel of  $T$  consists of all matrices of the form  $\begin{bmatrix} 0 & b \\ c & d \end{bmatrix}$ .

- (b) Let  $A$  and  $B$  be any  $2 \times 2$  matrices and let  $k$  be any real number. We have  
 $T(kA) = O_{2 \times 2} = kO_{2 \times 2} = kT(A)$  and  $T(A+B) = O_{2 \times 2} = O_{2 \times 2} + O_{2 \times 2} = T(A) + T(B)$   
 therefore  $T$  is a linear transformation.

The kernel of  $T$  is  $M_{22}$ .

- (c) Let  $A$  and  $B$  be any  $2 \times 2$  matrices and let  $k$  be any real number. We have  
 $T(kA) = c(kA) = k(cA) = kT(A)$  and  $T(A+B) = c(A+B) = cA + cB = T(A) + T(B)$   
 therefore  $T$  is a linear transformation.

The kernel of  $T$  consists of all matrices  $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$  such that

$$T\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right) = \begin{bmatrix} ca_{11} & ca_{12} \\ ca_{21} & ca_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

If  $c=0$  then  $\ker(T) = M_{22}$ , otherwise  $\ker(T) = \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\}$ .

3. For  $\mathbf{u} \neq \mathbf{0}$ ,  $T(-1\mathbf{u}) = \|- \mathbf{u}\| = \|\mathbf{u}\| = T(\mathbf{u}) \neq -1T(\mathbf{u})$ , so the mapping is not a linear transformation.
4. Let  $\mathbf{u}$  and  $\mathbf{v}$  be any vectors in  $R^3$  and let  $k$  be any real number. By properties of cross product listed in Theorem 3.5.2, we have  
 $T(k\mathbf{u}) = (k\mathbf{u}) \times \mathbf{v}_0 = k(\mathbf{u} \times \mathbf{v}_0) = kT(\mathbf{u})$  and  
 $T(\mathbf{u} + \mathbf{v}) = (\mathbf{u} + \mathbf{v}) \times \mathbf{v}_0 = (\mathbf{u} \times \mathbf{v}_0) + (\mathbf{v} \times \mathbf{v}_0) = T(\mathbf{u}) + T(\mathbf{v})$   
 therefore  $T$  is a linear transformation.  
 If  $\mathbf{v}_0 = \mathbf{0}$  then  $\ker(T) = R^3$ . Otherwise  $\ker(T) = \text{span}\{\mathbf{v}_0\}$ .
5. Let  $A_1$  and  $A_2$  be any  $2 \times 2$  matrices and let  $k$  be any real number. We have  
 $T(kA_1) = (kA_1)B = k(A_1B) = kT(A_1)$  and  
 $T(A_1 + A_2) = (A_1 + A_2)B = A_1B + A_2B = T(A_1) + T(A_2)$  therefore  $T$  is a linear transformation.  
 The kernel of  $T$  consists of all  $2 \times 2$  matrices whose rows are orthogonal to all columns of  $B$ .

$$\begin{aligned}
6. \quad (a) \quad T\left(k\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) &= T\left(\begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}\right) = 3ka - 4kb + kc - kd = k(3a - 4b + c - d) = kT\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) \\
T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}\right) &= T\left(\begin{bmatrix} a+a' & b+b' \\ c+c' & d+d' \end{bmatrix}\right) = 3(a+a') - 4(b+b') + (c+c') - (d+d') \\
&= (3a - 4b + c - d) + (3a' - 4b' + c' - d') = T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) + T\left(\begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}\right)
\end{aligned}$$

therefore  $T$  is a linear transformation.

The kernel of  $T$  consists of all matrices  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  for which  $3a - 4b + c - d = 0$ .

$$\begin{aligned}
(b) \quad T\left(2\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}\right) &= T\left(\begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix}\right) = 16 \text{ does not equal } 2T\left(\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}\right) = 2(4) = 8 \text{ therefore } T \text{ is not a linear} \\
&\text{transformation.}
\end{aligned}$$

$$7. \quad \text{Let } p(x) = a_0 + a_1x + a_2x^2 \text{ and } q(x) = b_0 + b_1x + b_2x^2.$$

$$\begin{aligned}
(a) \quad T(kp(x)) &= ka_0 + ka_1(x+1) + ka_2(x+1)^2 = kT(p(x)) \\
T(p(x) + q(x)) &= a_0 + b_0 + (a_1 + b_1)(x+1) + (a_2 + b_2)(x+1)^2 \\
&= a_0 + a_1(x+1) + a_2(x+1)^2 + b_0 + b_1(x+1) + b_2(x+1)^2 = T(p(x)) + T(q(x))
\end{aligned}$$

Thus  $T$  is a linear transformation.

The kernel of  $T$  consists of all polynomials  $a_0 + a_1x + a_2x^2$  such that

$$T(a_0 + a_1x + a_2x^2) = a_0 + a_1(x+1) + a_2(x+1)^2 = 0. \text{ This equality requires that}$$

$$a_0 = a_1 = a_2 = 0 \text{ therefore } \ker(T) = \{0\}.$$

$$\begin{aligned}
(b) \quad T(kp(x)) &= T(ka_0 + ka_1x + ka_2x^2) = (ka_0 + 1) + (ka_1 + 1)x + (ka_2 + 1)x^2 \neq kT(p(x)) \text{ so} \\
&T \text{ is not a linear transformation.}
\end{aligned}$$

$$8. \quad (a) \quad \text{Since } T \text{ maps the zero function } f(x) = 0 \text{ to } g(x) = 1, \text{ by Theorem 8.1.1(a), } T \text{ is not a linear transformation.}$$

$$\begin{aligned}
(b) \quad \text{Let } f \text{ and } g \text{ be any functions in } F(-\infty, \infty) \text{ and let } k \text{ be any real number. We have} \\
T(kf(x)) &= kf(x+1) = kT(f(x)) \text{ and } T(f(x) + g(x)) = f(x+1) + g(x+1) = T(f(x)) + T(g(x)) \\
&\text{therefore } T \text{ is a linear transformation.}
\end{aligned}$$

The kernel of  $T$  contains only the zero function.

$$\begin{aligned}
9. \quad T(k(a_0, a_1, a_2, \dots, a_n, \dots)) &= T(ka_0, ka_1, ka_2, \dots, ka_n, \dots) = (0, ka_0, ka_1, \dots, ka_n, \dots) \\
&= k(0, a_0, a_1, \dots, a_n, \dots) = kT(a_0, a_1, a_2, \dots, a_n, \dots) \\
T((a_0, a_1, a_2, \dots, a_n, \dots) + (b_0, b_1, b_2, \dots, b_n, \dots)) &= T(a_0 + b_0, a_1 + b_1, a_2 + b_2, \dots, a_n + b_n, \dots)
\end{aligned}$$

$$= (0, a_0 + b_0, a_1 + b_1, \dots, a_n + b_n, \dots) = (0, a_0, a_1, \dots, a_n, 0) + (0, b_0, b_1, \dots, b_n, 0)$$

$$= T(a_0, a_1, a_2, \dots, a_n, \dots) + T(b_0, b_1, b_2, \dots, b_n, \dots) \text{ therefore } T \text{ is a linear transformation.}$$
 The kernel of  $T$  contains only  $(0, 0, 0, \dots)$ .

10. (a)  $T(x^2) = (x)(x^2) = x^3 \neq 0$  therefore  $x^2$  is not in  $\ker(T)$ .
- (b)  $T(0) = (x)(0) = 0$  therefore  $0$  is in  $\ker(T)$ .
- (c)  $T(1+x) = (x)(1+x) = x + x^2 \neq 0$  therefore  $1+x$  is not in  $\ker(T)$ .
- (d)  $T(-x) = (x)(-x) = -x^2 \neq 0$  therefore  $-x$  is not in  $\ker(T)$ .
11. (a) Since  $x + x^2 = x(1+x)$ ,  $x + x^2$  is in  $R(T)$ .
- (b)  $1+x$  cannot be expressed in the form  $x p(x)$  for any polynomial  $p(x)$  therefore  $1+x$  is not in  $R(T)$ .
- (c)  $3 - x^2$  cannot be expressed in the form  $x p(x)$  for any polynomial  $p(x)$  therefore  $3 - x^2$  is not in  $R(T)$ .
- (d)  $-x = x(-1)$  therefore  $-x$  is in  $R(T)$ .
12. (a) The equation  $3\mathbf{v} = \mathbf{0}$  implies  $\mathbf{v} = \mathbf{0}$  therefore  $\ker(T) = \{\mathbf{0}\}$ .
- (b) Every vector  $\mathbf{w}$  in  $V$  is an image of  $\frac{1}{3}\mathbf{w}$  under  $T$ . Consequently,  $R(T) = V$ .
13. (a)  $\text{nullity}(T) = 5 - \text{rank}(T) = 2$
- (b)  $\dim(P_4) = 5$ , so  $\text{nullity}(T) = 5 - \text{rank}(T) = 4$
- (c) Since  $R(T) = R^3$ ,  $T$  has rank 3;  $\dim(M_{mn}) = mn$  so  $\text{nullity}(T) = mn - \text{rank}(T) = mn - 3$
- (d)  $\text{nullity}(T) = 4 - \text{rank}(T) = 1$
14. (a)  $\text{rank}(T) = 7 - \text{nullity}(T) = 5$
- (b)  $\text{rank}(T) = \dim(P_3) - \text{nullity}(T) = 4 - 1 = 3$
- (c) Since  $\ker(T) = P_5$ ,  $T$  has nullity 6;  $\text{rank}(T) = \dim(P_5) - \text{nullity}(T) = 6 - 6 = 0$
- (d)  $\text{rank}(T) = \dim(P_n) - \text{nullity}(T) = n + 1 - 3 = n - 2$
15. (a)  $T\left(\begin{bmatrix} 1 & 2 \\ -4 & 3 \end{bmatrix}\right) = 3\begin{bmatrix} 1 & 2 \\ -4 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ -12 & 9 \end{bmatrix}$

- (b) The only  $2 \times 2$  matrix  $A$  such that  $3A = 0$  is the zero matrix. Consequently,  $\ker(T) = \{0\}$  so the nullity of  $T$  is 0.

By Theorem 8.1.4,  $\text{rank}(T) = \dim(M_{22}) - \text{nullity}(T) = 4 - 0 = 4$ .

16. (a)  $T(1 + 4x + 8x^2) = \frac{1}{4} + x + 2x^2$

- (b) The only polynomial  $p$  in  $P_2$  such that  $\frac{1}{4}p = 0$  is the zero polynomial. Consequently,  $\ker(T) = \{0\}$  so the nullity of  $T$  is 0.

By Theorem 8.1.4,  $\text{rank}(T) = \dim(P_2) - \text{nullity}(T) = 3 - 0 = 3$ .

17. (a)  $T(x^2) = ((-1)^2, 0^2, 1^2) = (1, 0, 1)$

- (b) The kernel of  $T$  consists of all polynomials  $p(x) = a_0 + a_1x + a_2x^2$  such that  $(p(-1), p(0), p(1)) = (a_0 - a_1 + a_2, a_0, a_0 + a_1 + a_2) = (0, 0, 0)$ .

Equating the corresponding components we obtain a linear system

$$\begin{array}{rrcr} a_0 & - & a_1 & + & a_2 & = & 0 \\ a_0 & & & & & = & 0 \\ a_0 & + & a_1 & + & a_2 & = & 0 \end{array}$$

The reduced row echelon form of the coefficient matrix of this system is  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  hence the

system has a unique solution  $a_0 = a_1 = a_2 = 0$ . We conclude that  $\ker(T) = \{0\}$ .

- (c) It follows from the solution of part (b) that  $\text{nullity}(T) = 0$ .

By Theorem 8.1.4,  $\text{rank}(T) = \dim(P_2) - \text{nullity}(T) = 3 - 0 = 3$ . Consequently,  $R(T) = R^3$ .

18. (a)  $T(1 + \sin x + \cos x) = (1 + \sin 0 + \cos 0, 1 + \sin \pi + \cos \pi, 1 + \sin(2\pi) + \cos(2\pi)) = (2, 0, 2)$

- (b) The kernel of  $T$  consists of all functions  $f(x) = c_1 + c_2 \sin x + c_3 \cos x$  such that  $(f(0), f(\pi), f(2\pi))$

$$= (c_1 + c_2 \sin 0 + c_3 \cos 0, c_1 + c_2 \sin \pi + c_3 \cos \pi, c_1 + c_2 \sin(2\pi) + c_3 \cos(2\pi))$$

$$= (c_1 + c_3, c_1 - c_3, c_1 + c_3) = (0, 0, 0)$$

Equating the corresponding components yields  $c_1 = c_3 = 0$ . Since  $c_2$  is an arbitrary real number, we conclude that  $\ker(T) = \text{span}\{\sin x\}$ .

- (c)  $T(c_1 + c_2 \sin x + c_3 \cos x) = (c_1 + c_3, c_1 - c_3, c_1 + c_3) = c_1(1, 1, 1) + c_3(1, -1, 1)$ .

Consequently,  $R(T) = \text{span}\{(1, 1, 1), (1, -1, 1)\}$ .

19. For  $\mathbf{x} = (x_1, x_2) = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2$ , we have  $(x_1, x_2) = c_1(1, 1) + c_2(1, 0) = (c_1 + c_2, c_1)$  or

$$\begin{array}{rcl} c_1 + c_2 & = & x_1 \\ c_1 & = & x_2 \end{array}$$

which has the solution  $c_1 = x_2$ ,  $c_2 = x_1 - x_2$ .

$$(x_1, x_2) = x_2(1, 1) + (x_1 - x_2)(1, 0) = x_2 \mathbf{v}_1 + (x_1 - x_2) \mathbf{v}_2 \text{ and}$$

$$\begin{aligned} T(x_1, x_2) &= x_2 T(\mathbf{v}_1) + (x_1 - x_2) T(\mathbf{v}_2) = x_2(1, -2) + (x_1 - x_2)(-4, 1) = (-4x_1 + 5x_2, x_1 - 3x_2) \\ T(5, -3) &= (-20 - 15, 5 + 9) = (-35, 14). \end{aligned}$$

20. We begin by expressing  $(x_1, x_2)$  as a linear combination of the basis vectors  $(-2, 1)$  and  $(1, 3)$ :

$$(x_1, x_2) = c_1(-2, 1) + c_2(1, 3)$$

Equating the corresponding components we obtain a linear system

$$\begin{array}{rcl} -2c_1 + c_2 & = & x_1 \\ c_1 + 3c_2 & = & x_2 \end{array}$$

which yields  $c_1 = -\frac{3}{7}x_1 + \frac{1}{7}x_2$ ,  $c_2 = \frac{1}{7}x_1 + \frac{2}{7}x_2$ , allowing us to write

$$(x_1, x_2) = \left(-\frac{3}{7}x_1 + \frac{1}{7}x_2\right)(-2, 1) + \left(\frac{1}{7}x_1 + \frac{2}{7}x_2\right)(1, 3) \text{ and}$$

$$T(x_1, x_2) = \left(-\frac{3}{7}x_1 + \frac{1}{7}x_2\right)T(-2, 1) + \left(\frac{1}{7}x_1 + \frac{2}{7}x_2\right)T(1, 3)$$

$$= \left(-\frac{3}{7}x_1 + \frac{1}{7}x_2\right)(-1, 2, 0) + \left(\frac{1}{7}x_1 + \frac{2}{7}x_2\right)(0, -3, 5) = \left(\frac{3}{7}x_1 - \frac{1}{7}x_2, -\frac{9}{7}x_1 - \frac{4}{7}x_2, \frac{5}{7}x_1 + \frac{10}{7}x_2\right) \text{ so that}$$

$$T(2, -3) = \left(\frac{9}{7}, -\frac{6}{7}, -\frac{20}{7}\right).$$

21. For  $\mathbf{x} = (x_1, x_2, x_3) = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$ , we have

$$(x_1, x_2, x_3) = c_1(1, 1, 1) + c_2(1, 1, 0) + c_3(1, 0, 0) = (c_1 + c_2 + c_3, c_1 + c_2, c_1) \text{ or}$$

$$\begin{array}{rcl} c_1 + c_2 + c_3 & = & x_1 \\ c_1 + c_2 & = & x_2 \\ c_1 & = & x_3 \end{array}$$

which has the solution  $c_1 = x_3$ ,  $c_2 = x_2 - x_3$ ,  $c_3 = x_1 - (x_2 - x_3) - x_3 = x_1 - x_2$ .

$$(x_1, x_2, x_3) = x_3 \mathbf{v}_1 + (x_2 - x_3) \mathbf{v}_2 + (x_1 - x_2) \mathbf{v}_3$$

$$T(x_1, x_2, x_3) = x_3 T(\mathbf{v}_1) + (x_2 - x_3) T(\mathbf{v}_2) + (x_1 - x_2) T(\mathbf{v}_3)$$

$$= x_3(2, -1, 4) + (x_2 - x_3)(3, 0, 1) + (x_1 - x_2)(-1, 5, 1)$$

$$= (-x_1 + 4x_2 - x_3, 5x_1 - 5x_2 - x_3, x_1 + 3x_3)$$

$$T(2, 4, -1) = (-2 + 16 + 1, 10 - 20 + 1, 2 - 3) = (15, -9, -1)$$

22. We begin by expressing  $(x_1, x_2, x_3)$  as a linear combination of the basis vectors  $(1, 2, 1)$ ,  $(2, 9, 0)$ , and  $(3, 3, 4)$ :

$$(x_1, x_2, x_3) = c_1(1, 2, 1) + c_2(2, 9, 0) + c_3(3, 3, 4)$$

Equating the corresponding components we obtain a linear system

$$\begin{array}{rrcr} c_1 & + & 2c_2 & + & 3c_3 & = & x_1 \\ 2c_1 & + & 9c_2 & + & 3c_3 & = & x_2 \\ c_1 & & & + & 4c_3 & = & x_3 \end{array}$$

which yields  $c_1 = -36x_1 + 8x_2 + 21x_3$ ,  $c_2 = 5x_1 - x_2 - 3x_3$ ,  $c_3 = 9x_1 - 2x_2 - 5x_3$  allowing us to write

$$(x_1, x_2, x_3) = (-36x_1 + 8x_2 + 21x_3)(1, 2, 1) + (5x_1 - x_2 - 3x_3)(2, 9, 0) + (9x_1 - 2x_2 - 5x_3)(3, 3, 4) \text{ and}$$

$$T(x_1, x_2, x_3)$$

$$\begin{aligned} &= (-36x_1 + 8x_2 + 21x_3)T(1, 2, 1) + (5x_1 - x_2 - 3x_3)T(2, 9, 0) + (9x_1 - 2x_2 - 5x_3)T(3, 3, 4) \\ &= (-36x_1 + 8x_2 + 21x_3)(1, 0) + (5x_1 - x_2 - 3x_3)(-1, 1) + (9x_1 - 2x_2 - 5x_3)(0, 1) \\ &= (-41x_1 + 9x_2 + 24x_3, 14x_1 - 3x_2 - 8x_3) \end{aligned}$$

so that  $T(7, 13, 7) = (-2, 3)$ .

23. (a) The range of  $T(x) = Ax$  consists of all vectors  $(y_1, y_2, y_3)$  that are images of at least one vector

$$(x_1, x_2, x_3) \text{ under this transformation } \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ 5 & 6 & -4 \\ 7 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 6 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}. \text{ Since the}$$

reduced row echelon form of  $A$  is  $\begin{bmatrix} 1 & 0 & \frac{14}{11} \\ 0 & 1 & -\frac{19}{11} \\ 0 & 0 & 0 \end{bmatrix}$ , by Theorem 4.8.5 the vectors  $\begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$  and  $\begin{bmatrix} -1 \\ 6 \\ 4 \end{bmatrix}$  are

linearly independent, and they span  $R(T)$ . We conclude that  $\begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$  and  $\begin{bmatrix} -1 \\ 6 \\ 4 \end{bmatrix}$  form a basis for  $R(T)$ .

- (b) The kernel of  $T$  consists of all vectors  $(x_1, x_2, x_3)$  such that  $\begin{bmatrix} 1 & -1 & 3 \\ 5 & 6 & -4 \\ 7 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ . Based on the

reduced row echelon form of  $A$  obtained above, the general solution is  $x_1 = -\frac{14}{11}t$ ,  $x_2 = \frac{19}{11}t$ ,  $x_3 = t$ .

Therefore a basis for  $\ker(T)$  is formed by the vector  $(-14, 19, 11)$ .

- (c) From part (a),  $\text{rank}(T) = \dim(R(T)) = 2$ . From part (b),  $\text{nullity}(T) = \dim(\ker(T)) = 1$ .

- (d) Based on the reduced row echelon form of  $A$  obtained above,  $\text{rank}(A) = 2$  and  $\text{nullity}(A) = 1$ .

24. (a) The range of  $T(x) = Ax$  consists of all vectors  $(y_1, y_2, y_3)$  that are images of at least one vector

$$(x_1, x_2, x_3) \text{ under this transformation } \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 20 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 2 \\ 4 \\ 20 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}. \text{ Since the}$$

$$\text{reduced row echelon form of } A \text{ is } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \text{ by Theorem 4.8.5 the vectors } \begin{bmatrix} 2 \\ 4 \\ 20 \end{bmatrix} \text{ and } \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix} \text{ are}$$

linearly independent, and they span  $R(T)$ . We conclude that  $\begin{bmatrix} 2 \\ 4 \\ 20 \end{bmatrix}$  and  $\begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$  form a basis for  $R(T)$ .

- (b) The kernel of  $T$  consists of all vectors  $(x_1, x_2, x_3)$  such that  $\begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 20 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ . Based on the

reduced row echelon form of  $A$  obtained above, the general solution is  $x_1 = 0$ ,  $x_2 = t$ ,

$x_3 = 0$ . Therefore a basis for  $\ker(T)$  is formed by the vector  $(0, 1, 0)$ .

- (c) From part (a),  $\text{rank}(T) = \dim(R(T)) = 2$ . From part (b),  $\text{nullity}(T) = \dim(\ker(T)) = 1$ .

- (d) Based on the reduced row echelon form of  $A$  obtained above,  $\text{rank}(A) = 2$  and  $\text{nullity}(A) = 1$ .

25. The kernel of  $T_A$  consists of all vectors  $(x_1, x_2, x_3, x_4)$  such that  $\begin{bmatrix} 1 & 2 & -1 & -2 \\ -3 & 1 & 3 & 4 \\ -3 & 8 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ . Since the

$$\text{reduced row echelon form of } A \text{ is } \begin{bmatrix} 1 & 0 & 0 & -\frac{10}{7} \\ 0 & 1 & 0 & -\frac{2}{7} \\ 0 & 0 & 1 & 0 \end{bmatrix}, \text{ the general solution is } x_1 = \frac{10}{7}t, x_2 = \frac{2}{7}t,$$

$x_3 = 0$ ,  $x_4 = t$ . Therefore a basis for  $\ker(T_A)$  is formed by the vector  $(10, 2, 0, 7)$ .

The range of  $T_A(\mathbf{x}) = A\mathbf{x}$  consists of all vectors  $(y_1, y_2, y_3)$  that are images of at least one vector

$(x_1, x_2, x_3, x_4)$  under this transformation

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 & -2 \\ -3 & 1 & 3 & 4 \\ -3 & 8 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -3 \\ -3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 8 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 4 \\ 2 \end{bmatrix}. \text{ Based on the reduced row echelon}$$

form of  $A$  obtained above, by Theorem 4.8.5 the vectors  $\begin{bmatrix} 1 \\ -3 \\ -3 \end{bmatrix}$ ,  $\begin{bmatrix} 2 \\ 1 \\ 8 \end{bmatrix}$ , and  $\begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix}$  are linearly independent,

and they span  $R(T_A)$ . We conclude that  $\begin{bmatrix} 1 \\ -3 \\ -3 \end{bmatrix}$ ,  $\begin{bmatrix} 2 \\ 1 \\ 8 \end{bmatrix}$ , and  $\begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix}$  form a basis for  $R(T_A)$ .

26. The kernel of  $T_A$  consists of all vectors  $(x_1, x_2, x_3, x_4)$  such that  $\begin{bmatrix} 1 & 1 & 0 & 1 \\ -2 & 4 & 2 & 2 \\ -1 & 8 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ . Since the

reduced row echelon form of  $A$  is  $\begin{bmatrix} 1 & 0 & -\frac{1}{3} & \frac{1}{3} \\ 0 & 1 & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}$ , the general solution is  $x_1 = \frac{1}{3}s - \frac{1}{3}t$ ,

$x_2 = -\frac{1}{3}s - \frac{2}{3}t$ ,  $x_3 = s$ ,  $x_4 = t$ . In vector form,  $(x_1, x_2, x_3, x_4) = s(\frac{1}{3}, -\frac{1}{3}, 1, 0) + t(-\frac{1}{3}, -\frac{2}{3}, 0, 1)$ . Therefore a basis for  $\ker(T_A)$  is formed by the vectors  $(1, -1, 3, 0)$  and  $(-1, -2, 0, 3)$ .

The range of  $T_A(\mathbf{x}) = A\mathbf{x}$  consists of all vectors  $(y_1, y_2, y_3)$  that are images of at least one vector  $(x_1, x_2, x_3, x_4)$  under this transformation

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ -2 & 4 & 2 & 2 \\ -1 & 8 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}. \text{ Based on the reduced row echelon form of}$$

$A$  obtained above, by Theorem 4.8.5 the vectors  $\begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$  and  $\begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix}$  are linearly independent, and they span

$R(T_A)$ . We conclude that  $\begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$  and  $\begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix}$  form a basis for  $R(T_A)$ .

27. (a)  $T(k(a_0 + a_1x + a_2x^2 + a_3x^3)) = T(ka_0 + ka_1x + ka_2x^2 + ka_3x^3) = 5ka_0 + ka_3x^2$   
 $= k(5a_0 + a_3x^2) = kT(a_0 + a_1x + a_2x^2 + a_3x^3)$   
 $T(a_0 + a_1x + a_2x^2 + a_3x^3 + b_0 + b_1x + b_2x^2 + b_3x^3)$   
 $= T((a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + (a_3 + b_3)x^3) = 5(a_0 + b_0) + (a_3 + b_3)x^2$   
 $= (5a_0 + a_3x^2) + (5b_0 + b_3x^2) = T(a_0 + a_1x + a_2x^2 + a_3x^3) + T(b_0 + b_1x + b_2x^2 + b_3x^3)$   
 therefore  $T$  is linear.

- (b) The kernel of  $T$  consists of all polynomials  $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$  such that

$$T(a_0 + a_1x + a_2x^2 + a_3x^3) = 5a_0 + a_3x^2 = 0, \text{ which requires that } a_0 = a_3 = 0.$$

Therefore every vector in  $\ker(T)$  can be written in the form  $a_1x + a_2x^2$ , i.e.,

$\ker(T) = \text{span}\{x, x^2\}$ . The set  $\{x, x^2\}$  is linearly independent since neither polynomial is a scalar multiple of the other one. We conclude that  $\{x, x^2\}$  is a basis for  $\ker(T)$ .

- (c)  $T(a_0 + a_1x + a_2x^2 + a_3x^3) = 5a_0 + a_3x^2$  so  $R(T) = \text{span}\{5, x^2\}$ . The set  $\{5, x^2\}$  is linearly independent since neither polynomial is a scalar multiple of the other one.

We conclude that  $\{5, x^2\}$  is a basis for  $R(T)$ .

28. (a) 
$$\begin{aligned} T(k(a_0 + a_1x + a_2x^2)) &= T(ka_0 + ka_1x + ka_2x^2) = 3ka_0 + ka_1x + (ka_0 + ka_1)x^2 \\ &= k(3a_0 + a_1x + (a_0 + a_1)x^2) = kT(a_0 + a_1x + a_2x^2) \\ T(a_0 + a_1x + a_2x^2 + b_0 + b_1x + b_2x^2) &= T((a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2) = 3(a_0 + b_0) + (a_1 + b_1)x + (a_0 + b_0 + a_1 + b_1)x^2 \\ &= (3a_0 + a_1x + (a_0 + a_1)x^2) + (3b_0 + b_1x + (b_0 + b_1)x^2) \\ &= T(a_0 + a_1x + a_2x^2) + T(b_0 + b_1x + b_2x^2) \end{aligned}$$
 therefore  $T$  is linear.

- (b) The kernel of  $T$  consists of all polynomials  $p(x) = a_0 + a_1x + a_2x^2$  such that

$$T(a_0 + a_1x + a_2x^2) = 3a_0 + a_1x + (a_0 + a_1)x^2 = 0, \text{ which requires that } a_0 = a_1 = 0.$$

Therefore every vector in  $\ker(T)$  can be written in the form  $a_2x^2$ , i.e.,  $\ker(T) = \text{span}\{x^2\}$ .

The set  $\{x^2\}$  is linearly independent since  $x^2$  is not the zero polynomial.

We conclude that  $\{x^2\}$  is a basis for  $\ker(T)$ .

- (c)  $T(a_0 + a_1x + a_2x^2) = 3a_0 + a_1x + (a_0 + a_1)x^2 = a_0(3 + x^2) + a_1(x + x^2)$   
so  $R(T) = \text{span}\{3 + x^2, x + x^2\}$ .

The set  $\{3 + x^2, x + x^2\}$  is linearly independent since neither polynomial is a scalar multiple of the other one. We conclude that  $\{3 + x^2, x + x^2\}$  is a basis for  $R(T)$ .

29. (a) If  $p'(x) = 0$ , then  $p(x)$  is a constant, so  $\ker(D)$  consists of all constant polynomials.

- (b) The kernel of  $J$  contains all polynomials  $a_0 + a_1x$  such that  $\int_{-1}^1 (a_0 + a_1x) dx = 0$ . By integration, this condition yields  $\left(a_0x + \frac{a_1x^2}{2}\right)\Big|_{-1}^1 = 0$ , i.e.,  $a_0 + \frac{a_1}{2} + a_0 - \frac{a_1}{2} = 0$ , or equivalently,  $a_0 = 0$ .

The kernel consists of all polynomials of the form  $a_1x$ .

30. For any functions  $f$  and  $g$  in  $C[a, b]$  and for any real number  $k$  we have

$$T(k\mathbf{f}) = 5kf(x) + 3\int_a^x kf(t) dt = k\left(5f(x) + \int_a^x f(t) dt\right) = kT(\mathbf{f}) \text{ and}$$

$$\begin{aligned} T(\mathbf{f} + \mathbf{g}) &= 5(f(x) + g(x)) + 3\int_a^x (f(t) + g(t)) dt = \left(5f(x) + 3\int_a^x f(t) dt\right) + \left(5g(x) + 3\int_a^x g(t) dt\right) \\ &= T(\mathbf{f}) + T(\mathbf{g}) \end{aligned}$$

This shows that  $T$  is a linear operator.

31. (a) If  $f^4(x) = 0$ , then  $f'''(x) = a$  for some constant  $a$ . Applying Fundamental Theorem of Calculus, we obtain  $f''(x) = ax + b$ , then  $f'(x) = \frac{a}{2}x^2 + bx + c$ , and  $f(x) = \frac{a}{6}x^3 + \frac{b}{2}x^2 + cx + d$  for constants  $b$ ,  $c$ , and  $d$ . We conclude that  $P_3$  is the kernel of  $T(f(x)) = f^{(4)}(x)$ .

- (b) By similar reasoning,  $T(f(x)) = f^{(n+1)}(x)$  has  $\ker(T) = P_n$ .

32. Since  $R(T) = R$ , it follows from Theorem 8.1.4 that  $\ker(T)$  has dimension

$$\dim(M_{nn}) - \dim(R) = n^2 - 1.$$

33. (a)  $R(T)$  must be a subspace of  $R^3$ , thus the possibilities are a line through the origin, a plane through the origin, the origin only, or all of  $R^3$ .

- (b) The origin, a line through the origin, a plane through the origin, or the entire space  $R^3$ .

34. (a) For all polynomials  $p(x)$  and  $q(x)$  in  $P_n$  and for every scalar  $k$  we have

$$T(kp(x)) = kp(x+1) = kT(p(x)) \text{ and}$$

$$T(p(x) + q(x)) = p(x+1) + q(x+1) = T(p(x)) + T(q(x)) \text{ therefore } T \text{ is linear.}$$

- (b) Since  $T$  maps the zero polynomial  $p(x) = 0$  to  $q(x) = 1$ , by Theorem 8.1.1(a),  $T$  is not a linear transformation.

35.  $T(2\mathbf{v}_1 - 3\mathbf{v}_2 + 4\mathbf{v}_3) = 2T(\mathbf{v}_1) - 3T(\mathbf{v}_2) + 4T(\mathbf{v}_3) = (2, -2, 4) - (0, 9, 6) + (-12, 4, 8)$   
 $= (-10, -7, 6)$

36. Let  $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$  be any vector in  $V$ . Then

$$T(\mathbf{v}) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \cdots + c_nT(\mathbf{v}_n) = c_1\mathbf{0} + c_2\mathbf{0} + \cdots + c_n\mathbf{0} = \mathbf{0}$$

Since  $\mathbf{v}$  was an arbitrary vector in  $V$ ,  $T$  must be the zero transformation.

37. Let  $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$  be any vector in  $V$ . Then

$$T(\mathbf{v}) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \cdots + c_nT(\mathbf{v}_n) = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{v}$$

Since  $\mathbf{v}$  was an arbitrary vector in  $V$ ,  $T$  must be the identity operator.

38. For every vector  $\mathbf{v}$  in  $V$ , there are unique scalars  $c_1, c_2, \dots, c_n$  such that  $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$ .

A linear transformation with the desired properties can be defined by

$$T(\mathbf{v}) = T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n) = c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + \cdots + c_n\mathbf{w}_n$$

39.  $T(kp(x)) = kp(q_0(x)) = kT(p(x))$

$$T(p_1(x) + p_2(x)) = p_1(q_0(x)) + p_2(q_0(x)) = T(p_1(x)) + T(p_2(x))$$

### True-False Exercises

- (a) True.  $c_1 = k, c_2 = 0$  gives the homogeneity property and  $c_1 = c_2 = 1$  gives the additivity property.
- (b) False. Every linear transformation will have  $T(-\mathbf{v}) = -T(\mathbf{v})$ .
- (c) True. Only the zero transformation has this property.
- (d) False.  $T(\mathbf{0}) = \mathbf{v}_0 + \mathbf{0} = \mathbf{v}_0 \neq \mathbf{0}$ , so  $T$  is not a linear transformation.
- (e) True. This follows from part (a) of Theorem 8.1.3.
- (f) True. This follows from part (b) of Theorem 8.1.3.
- (g) False.  $T$  does not necessarily have rank 4.
- (h) False.  $\det(A + B) \neq \det(A) + \det(B)$  in general.
- (i) False.  $\text{nullity}(T) = \text{rank}(T) = 2$

## 8.2 Compositions and Inverse Transformations

- 1. (a) Not one-to-one (maps distinct vectors with the same  $x$  components into the same vector).
- (b) One-to-one (distinct vectors that are reflected have distinct images).
- (c) One-to-one (distinct vectors that are reflected have distinct images).
- 2. (a) One-to-one (distinct vectors that are rotated have distinct images).
- (b) One-to-one (maps distinct vectors with the same  $x$  and  $y$  components into the same vector).
- (c) Not one-to-one (distinct vectors that are contracted have distinct images).
- 3. (a) By inspection,  $\ker(T) = \{\mathbf{0}\}$ , so  $T$  is one-to-one.
- (b) By inspection,  $\ker(T) = \{\mathbf{0}\}$ , so  $T$  is one-to-one.

(c)  $(x, y, z)$  is in  $\ker(T)$  if both  $x + y + z = 0$  and  $x - y - z = 0$ , which is  $x = 0$  and  $y + z = 0$ .  
Thus,  $\ker(T) = \text{span}\{(0, 1, 1)\}$  and  $T$  is not one-to-one.

4. (a)  $(x, y)$  is in  $\ker(T)$  if  $x - y = 0$  or  $x = y$ , so  $\ker(T) = \text{span}\{(1, 1)\}$  and  $T$  is not one-to-one.

(b)  $T(x, y) = \mathbf{0}$  if  $2x + 3y = 0$  or  $x = -\frac{3}{2}y$  so  $\ker(T) = \text{span}\{(-\frac{3}{2}, 1)\}$  and  $T$  is not one-to-one.

(c)  $(x, y)$  is in  $\ker(T)$  only if  $x + y = 0$  and  $x - y = 0$ , so  $\ker(T) = \{\mathbf{0}\}$  and  $T$  is one-to-one.

5. (a) The reduced row echelon form of  $A$  is  $\begin{bmatrix} 1 & -2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ , so  $\text{nullity}(A) = 1$ .

Multiplication by  $A$  is not one-to-one.

(b) The reduced row echelon form of  $A$  is  $\begin{bmatrix} 1 & 0 & 0 & 30 \\ 0 & 1 & 0 & -10 \\ 0 & 0 & 1 & 7 \end{bmatrix}$ , so  $\text{nullity}(A) = 1$ .

Multiplication by  $A$  is not one-to-one.

6. (a) The reduced row echelon form of  $A$  is  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ , so  $\text{nullity}(A) = 0$ .

Multiplication by  $A$  is one-to-one.

(b) The reduced row echelon form of  $A$  is  $\begin{bmatrix} 1 & 0 & 12 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ , so  $\text{nullity}(A) = 1$ .

Multiplication by  $A$  is not one-to-one.

7. (a) Since  $\text{nullity}(T) = 0$ ,  $T$  is one-to-one.

(b)  $\text{nullity}(T) = \dim(V) - \text{rank}(T) = 0$  therefore  $T$  is one-to-one.

(c) Since  $\text{rank}(T) \leq \dim(W) < \dim(V)$ , we have  $\text{nullity}(T) = \dim(V) - \text{rank}(T) > 0$ . We conclude that  $T$  is not one-to-one.

8. (a) Since  $\text{nullity}(T) = 0$ ,  $T$  is one-to-one;  $\text{rank}(T) = \dim(V) - \text{nullity}(T) = \dim(V)$  so  $T$  is onto.

(b)  $\text{nullity}(T) = \dim(V) - \text{rank}(T) > 0$  so  $T$  is not one-to-one;  $\text{rank}(T) < \dim(V)$  so  $T$  is not onto.

(c)  $R(T) = V$  so  $T$  is onto;  $\text{nullity}(T) = \dim(V) - \text{rank}(T) = 0$  so  $T$  is one-to-one.

9. For example,  $T(1-x^2) = (1-(-1)^2, 1-1^2) = (0, 0)$ .

The transformation is onto since for any real numbers  $a$  and  $b$ , a polynomial  $p(x)$  in  $P_2$  can be found such that  $p(-1) = a$  and  $p(1) = b$ .

10. Setting  $a_0 + a_1(x+1) + a_2(x+1)^2 = 0$  implies  $a_0 = a_1 = a_2 = 0$ . Since  $\ker(T) = \{0\}$ , by Theorem 8.2.1,  $T$  is one-to-one.

$\text{rank}(T) = \dim(P_2) - \text{nullity}(T) = \dim(P_2)$  therefore  $T$  is onto.

11. No;  $T$  is not one-to-one because  $\ker(T) \neq \{0\}$  as  $T(\mathbf{a}) = \mathbf{a} \times \mathbf{a} = \mathbf{0}$ .

12. Since elementary matrices are invertible,  $EA = 0$  yields  $A = E^{-1}0 = 0$  therefore  $\ker(T) = \{0\}$ .

Consequently, by Theorem 8.2.1  $T$  is a one-to-one linear operator.

13. (a) The columns of  $A$  are linearly independent by inspection, so multiplication by  $A$  is one-to one by Theorem 8.2.3(a). Since  $A$  has only two columns, they do not span  $R^3$ . Therefore, by Theorem 8.2.3(b), multiplication by  $A$  is not onto.

(b) The columns of  $A$  are linearly dependent since they form a set of four vectors in  $R^3$  so multiplication by  $A$  is not one-to-one by Theorem 8.2.3(a). The first, third, and fourth columns of  $A$  are linearly

independent since  $\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = -1 \neq 0$  so these columns span  $R^3$ . By Theorem 8.2.3(b),

multiplication by  $A$  is onto.

(c) Since  $\det(A) = \det \begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} = 1 \neq 0$ , the columns of  $A$  are a basis of  $R^2$ . By Theorem 8.2.3,

multiplication by  $A$  is both one-to-one and onto.

(d) Since  $\det(a) = \det \begin{pmatrix} -2 & 1 & 0 \\ 6 & -3 & 1 \\ 8 & -4 & 3 \end{pmatrix} = 0$ , the columns of  $A$  are linearly dependent which implies they do

not span  $R^3$ . By Theorem 8.2.3, multiplication by  $A$  is neither one-to-one nor onto.

14. (a) The columns of  $A$  are linearly independent by inspection, so multiplication by  $A$  is one-to one by Theorem 8.2.3(a). Since  $A$  has only two columns, they do not span  $R^3$ . Therefore, by Theorem 8.2.3(b), multiplication by  $A$  is not onto.

(b) The columns of  $A$  are linearly dependent since they form a set of four vectors in  $R^3$  so multiplication by  $A$  is not one-to-one by Theorem 8.2.3(a). The second, third, and fourth columns of  $A$  are linearly

dependent since  $\det \begin{pmatrix} -3 & 1 & 1 \\ -6 & 0 & 2 \\ -9 & 1 & 3 \end{pmatrix} = 0$ . Also, first column is 3 times the third column and the second

column of  $A$  is  $-3$  times the first column. Therefore,  $A$  has fewer than three linearly independent columns so its columns do not span  $R^3$ . By Theorem 8.2.3(b), multiplication by  $A$  is not onto.

(c) Since  $\det(A) = \det\begin{bmatrix} 3 & -9 \\ -1 & 3 \end{bmatrix} = 0$ , the columns of  $A$  are linearly dependent and, therefore, do not span  $R^2$ . By Theorem 8.2.3, multiplication by  $A$  is neither one-to-one nor onto.

(d) Since  $\det(A) = \det\begin{bmatrix} 2 & 3 & 8 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} = 2 \neq 0$ , the columns of  $A$  are a basis for  $R^3$ . By Theorem 8.2.3, multiplication by  $A$  is both one-to-one and onto.

15. (a) The reflection about the  $x$ -axis in  $R^2$  is its own inverse.  
 (b) The rotation through an angle of  $-\pi/4$  in  $R^2$  (i.e., the clockwise rotation through an angle  $\pi/4$ ) is the desired inverse.
16. (a) The reflection about the  $yz$ -plane in  $R^3$  is its own inverse.  
 (b) The rotation through an angle of  $18^\circ$  about the  $z$ -axis is the desired inverse.
17. (a) The standard matrix of the reflection about the line  $y = x$  is  $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ .

$$\text{The inverse is } A^{-1} = \frac{1}{(0)(0) - (1)(1)} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = A.$$

- (b) The standard matrix of the rotation by angle  $\theta$  about the origin is  $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ .

The inverse is  $A^{-1} = \frac{1}{(\cos \theta)(\cos \theta) - (-\sin \theta)(-\sin \theta)} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix}$  which is a rotation by angle  $-\theta$  about the origin.

18. (a) The standard matrix of the reflection about the  $y$ -axis is  $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ .

$$\text{The inverse is } A^{-1} = \frac{1}{(-1)(1) - (0)(0)} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = A.$$

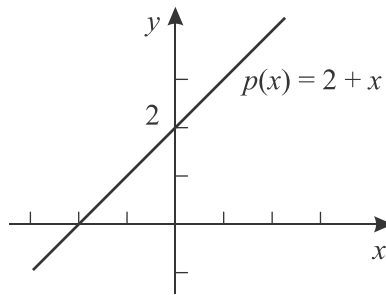
(Reflections about the  $x$ -axis can be treated analogously.)

- (b) The standard matrix of the reflection about the origin is  $A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ .

$$\text{The inverse is } A^{-1} = \frac{1}{(0)(0) - (-1)(-1)} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = A.$$

19. (a)  $T(1 - 2x) = (1 - 2(0), 1 - 2(1)) = (1, -1)$

- (b)  $T(kp(x)) = (kp(0), kp(1)) = k(p(0), p(1)) = kT(p(x));$   
 $T(p(x) + q(x)) = (p(0) + q(0), p(1) + q(1)) = (p(0), p(1)) + (q(0), q(1))$   
 $= T(p(x)) + T(q(x))$
- (c) Let  $p(x) = a_0 + a_1x$ , then  $T(p(x)) = (a_0, a_0 + a_1)$  so if  
 $T(p(x)) = (0, 0)$ , then  $a_0 = a_1 = 0$  and  $p$  is the zero polynomial, so  $\ker(T) = \{0\}$ .
- (d) Since  $T(p(x)) = (a_0, a_0 + a_1)$ , then  $T^{-1}(2, 3)$  has  $a_0 = 2$  and  $a_0 + a_1 = 3$  or  $a_1 = 1$ . Thus,  
 $T^{-1}(2, 3) = 2 + x$ .



20. (a)  $T$  is not one-to-one, e.g.,  $T(0, \dots, 0, 1) = T(0, \dots, 0, 2) = (0, \dots, 0, 0)$ .
- (b)  $\ker(T) = \{(0, \dots, 0)\}$  thus  $T$  is one-to-one;  
 $T^{-1}(x_1, x_2, \dots, x_n) = (x_n, x_{n-1}, \dots, x_2, x_1)$  (i.e.,  $T^{-1} = T$ )
- (c)  $\ker(T) = \{(0, \dots, 0)\}$  thus  $T$  is one-to-one;  $T^{-1}(x_1, x_2, \dots, x_n) = (x_n, x_1, \dots, x_{n-1})$
21. (a) For  $T$  to have an inverse, all the  $a_i$ 's must be nonzero since otherwise  $T$  would have a nonzero kernel.
- (b)  $T^{-1}(x_1, x_2, \dots, x_n) = (\frac{1}{a_1}x_1, \frac{1}{a_2}x_2, \dots, \frac{1}{a_n}x_n)$
22. Observe that  $\begin{bmatrix} 1 & 0 & 2 & 5 \\ 3 & 4 & 1 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  so that  $T_A(0, 0, -2, 1) = (1, 1)$ . Therefore, any vector  $\mathbf{v}$  in  $R^2$  for which  $T_A(\mathbf{v}) = (1, 1)$  can be expressed in vector form as  $\mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} + \mathbf{u}$  for some vector  $\mathbf{u}$  in the kernel of  $T_A$ . The reduced row echelon form of  $\begin{bmatrix} 1 & 0 & 2 & 5 \\ 3 & 4 & 1 & 3 \end{bmatrix}$  is  $\begin{bmatrix} 1 & 0 & 2 & 5 \\ 0 & 1 & -\frac{5}{4} & -3 \end{bmatrix}$  so any vector in the kernel may

be expressed parametrically as  $\mathbf{u} = t \begin{bmatrix} -2 \\ \frac{5}{4} \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -5 \\ 3 \\ 0 \\ 1 \end{bmatrix}$ . Therefore, any vector  $\mathbf{v}$  in  $R^2$  for which  $T_A(\mathbf{v}) = (1, 1)$

can be written parametrically as  $\mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} + t \begin{bmatrix} -2 \\ \frac{5}{4} \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -5 \\ 3 \\ 0 \\ 1 \end{bmatrix}$ .

23.  $(T_2 \circ T_1)(x, y) = T_2(2x, 3y) = (2x - 3y, 2x + 3y)$
24.  $(T_2 \circ T_1)(x, y) = T_2(2x, -3y, x + y) = (2x + 3y, -3y + x + y) = (2x + 3y, x - 2y)$
25.  $(T_2 \circ T_1)(a_0 + a_1x + a_2x^2) = T_2(T_1(a_0 + a_1x + a_2x^2)) = T_2(a_0 + a_1(x+1) + a_2(x+1)^2)$   
 $= x(a_0 + a_1(x+1) + a_2(x+1)^2) = a_0x + a_1x(x+1) + a_2x(x+1)^2$
26.  $(T_1 \circ T_2)(p(x)) = T_1(T_2(p(x))) = T_1(p(x+1)) = p((x+1)-1) = p(x)$   
 $(T_2 \circ T_1)(p(x)) = T_2(T_1(p(x))) = T_2(p(x-1)) = p((x-1)+1) = p(x)$
27. (a)  $(T_1 \circ T_2)(A) = T_1(A^T) = \text{tr} \left( \begin{bmatrix} a & c \\ b & d \end{bmatrix} \right) = a + d$
- (b)  $(T_2 \circ T_1)(A)$  does not exist because  $T_1(A)$  is not a  $2 \times 2$  matrix.
28. (a)  $(T_1 \circ T_2)(A) = T_1(A^T) = k \left( \begin{bmatrix} a & c \\ b & d \end{bmatrix} \right) = \begin{bmatrix} ka & kc \\ kb & kd \end{bmatrix}$
- (b)  $(T_2 \circ T_1)(A) = T_2 \left( \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix} \right) = \begin{bmatrix} ka & kc \\ kb & kd \end{bmatrix}$
29.  $(T_3 \circ T_2 \circ T_1)(x, y) = T_3(T_2(T_1(x, y))) = T_3(T_2(-2y, 3x, x - 2y)) = T_3(3x, x - 2y, -2y)$   
 $= (3x - 2y, x - 2y - (-2y)) = (3x - 2y, x)$
30.  $(T_3 \circ T_2 \circ T_1)(x, y) = T_3(T_2(T_1(x, y))) = T_3(T_2(x + y, y, -x)) = T_3(0, x + y + y - x, 3y)$   
 $= T_3(0, 2y, 3y) = (4y, 12y - 6y) = (4y, 6y)$
31. (a) Since  $T_1(p(x)) = xp(x)$ ,  $T_1^{-1}(p(x)) = \frac{1}{x}p(x)$ .  
 Since  $T_2(p(x)) = p(x+1)$ ,  $T_2^{-1}(p(x)) = p(x-1)$ .  
 $(T_1^{-1} \circ T_2^{-1})(p(x)) = T_1^{-1}(p(x-1)) = \frac{1}{x}p(x-1)$

(b) Since  $(T_2 \circ T_1)(p(x)) = T_2(T_1(p(x))) = T_2(xp(x)) = (x+1)p(x+1)$ , we have

$$(T_2 \circ T_1)\left(\left(T_1^{-1} \circ T_2^{-1}\right)(p(x))\right) = (T_2 \circ T_1)\left(\left(\frac{1}{x}\right)p(x-1)\right) = (x+1)\left(\frac{1}{x+1}\right)p(x-1+1) = p(x)$$

32. (a) Setting  $T_1(x, y) = (x+y, x-y) = (0, 0)$  and  $T_2(x, y) = (2x+y, x-2y) = (0, 0)$  yields the systems

$$\begin{array}{rcl} x + y & = & 0 \\ x - y & = & 0 \end{array} \quad \text{and} \quad \begin{array}{rcl} 2x + y & = & 0 \\ x - 2y & = & 0 \end{array}$$

each of which has only the trivial solution  $(x, y) = (0, 0)$  therefore  $\ker(T_1) = \ker(T_2) = \{(0, 0)\}$ . By Theorem 8.2.1,  $T_1$  and  $T_2$  are one-to-one.

(b) The standard matrices of  $T_1$  and  $T_2$  are

$$[T_1] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \text{ and } [T_2] = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}, \text{ respectively. Both matrices are invertible, and we have}$$

$$[T_1^{-1}] = [T_1]^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \text{ and } [T_2^{-1}] = [T_2]^{-1} = \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ \frac{1}{5} & -\frac{2}{5} \end{bmatrix} \text{ so that}$$

$$T_1^{-1}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = [T_1^{-1}]\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x + \frac{1}{2}y \\ \frac{1}{2}x - \frac{1}{2}y \end{bmatrix}, \text{ i.e. } T_1^{-1}(x, y) = \left(\frac{1}{2}x + \frac{1}{2}y, \frac{1}{2}x - \frac{1}{2}y\right) \text{ and}$$

$$T_2^{-1}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = [T_2^{-1}]\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ \frac{1}{5} & -\frac{2}{5} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{2}{5}x + \frac{1}{5}y \\ \frac{1}{5}x - \frac{2}{5}y \end{bmatrix}, \text{ i.e. } T_2^{-1}(x, y) = \left(\frac{2}{5}x + \frac{1}{5}y, \frac{1}{5}x - \frac{2}{5}y\right).$$

Since  $(T_2 \circ T_1)(x, y) = T_2(T_1(x, y)) = T_2(x+y, x-y) = (3x+y, -x+3y)$  we have

$$[T_2 \circ T_1] = \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix} \text{ which is an invertible matrix with } [T_2 \circ T_1]^{-1} = \begin{bmatrix} \frac{3}{10} & -\frac{1}{10} \\ \frac{1}{10} & \frac{3}{10} \end{bmatrix}.$$

$$\text{Therefore } (T_2 \circ T_1)^{-1}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = [T_2 \circ T_1]^{-1}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{3}{10} & -\frac{1}{10} \\ \frac{1}{10} & \frac{3}{10} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{3}{10}x - \frac{1}{10}y \\ \frac{1}{10}x + \frac{3}{10}y \end{bmatrix}, \text{ i.e.,}$$

$$(T_2 \circ T_1)^{-1}(x, y) = \left(\frac{3}{10}x - \frac{1}{10}y, \frac{1}{10}x + \frac{3}{10}y\right).$$

$$\begin{aligned} \text{(c) } (T_1^{-1} \circ T_2^{-1})(x, y) &= T_1^{-1}(T_2^{-1}(x, y)) = T_1^{-1}\left(\frac{2}{5}x + \frac{1}{5}y, \frac{1}{5}x - \frac{2}{5}y\right) \\ &= \left(\frac{1}{2}\left(\frac{2}{5}x + \frac{1}{5}y\right) + \frac{1}{2}\left(\frac{1}{5}x - \frac{2}{5}y\right), \frac{1}{2}\left(\frac{2}{5}x + \frac{1}{5}y\right) - \frac{1}{2}\left(\frac{1}{5}x - \frac{2}{5}y\right)\right) = \left(\frac{3}{10}x - \frac{1}{10}y, \frac{1}{10}x + \frac{3}{10}y\right) \\ &= (T_2 \circ T_1)^{-1}(x, y) \end{aligned}$$

33.  $T_2(\mathbf{v}) = \frac{1}{4}\mathbf{v}$ , then  $(T_1 \circ T_2)(\mathbf{v}) = T_1\left(\frac{1}{4}\mathbf{v}\right) = 4\left(\frac{1}{4}\mathbf{v}\right) = \mathbf{v}$  and  $(T_2 \circ T_1)(\mathbf{v}) = T_2(4\mathbf{v}) = \frac{1}{4}(4\mathbf{v}) = \mathbf{v}$ .

$$34. \text{ (a) } (T_2 \circ T_1)\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = T_2((a+b) + (c+d)x) = (a+b, c+d, a+b)$$

$$\text{(b) For example, } (T_2 \circ T_1)\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = (T_2 \circ T_1)\left(\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}\right) = (1, 1, 1).$$

(c) For example,  $(1, 2, 3)$  cannot be expressed in the form  $(a + b, c + d, a + b)$ .

35. By inspection,  $T(x, y, z) = (x, y, 0)$ . Then  $T(T(x, y, z)) = T(x, y, 0) = (x, y, 0) = T(x, y, z)$  or  $T \circ T = T$ .

36.  $T(kf) = kf(0) + 2kf'(0) + 3kf'(1) = kT(f)$

$$T(f + g) = f(0) + g(0) + 2(f'(0) + g'(0)) + 3(f'(1) + g'(1)) = T(f) + T(g)$$

Thus  $T$  is a linear transformation.

Let  $f = f(x) = x^2(x - 1)^2$ , then  $f'(x) = 2x(x - 1)(2x - 1)$  so  $f(0) = 0$ ,  $f'(0) = 0$ , and  $f'(1) = 0$ .

$T(f) = f(0) + 2f'(0) + 3f'(1) = 0$ , so  $\ker(T) \neq \{0\}$  and  $T$  is not one-to-one.

37. (a)  $D(kp(x)) = \frac{d}{dx}(kp(x)) = kp'(x) = kD(p(x))$

$$D(p(x) + q(x)) = \frac{d}{dx}(p(x) + q(x)) = p'(x) + q'(x) = D(p(x)) + D(q(x))$$

$$J(kp(x)) = \int_0^x kp(t) dt = k \int_0^x p(t) dt = kJ(p(x))$$

$$J(p(x) + q(x)) = \int_0^x (p(t) + q(t)) dt = \int_0^x p(t) dt + \int_0^x q(t) dt = J(p(x)) + J(q(x))$$

(b)  $D$  is not one-to-one (e.g.,  $D(x^2 + 3) = D(x^2) = 2x$ ) so it does not have an inverse.

(c) Yes, this can be accomplished by taking  $D: V \rightarrow P_{n-1}$  and  $J: P_{n-1} \rightarrow V$  where  $V$  is the set of all polynomials  $p(x)$  in  $P_n$  such that  $p(0) = 0$ .

38. (a)  $(J \circ D)(f) = J(D(x^2 + 3x + 2)) = J(2x + 3) = \int_0^x (2t + 3) dt = \left[ t^2 + 3t \right]_0^x = x^2 + 3x$

(b)  $(J \circ D)(f) = J(D(\sin x)) = J(\cos x) = \int_0^x (\cos t) dt = \sin t \Big|_0^x = \sin x$

39. The kernel of  $J$  contains all polynomials  $a_0 + a_1x$  such that  $\int_{-1}^1 (a_0 + a_1x) dx = 0$ . By integration, this condition yields  $\left( a_0x + \frac{a_1x^2}{2} \right) \Big|_{-1}^1 = 0$ , i.e.,  $a_0 + \frac{a_1}{2} + a_0 - \frac{a_1}{2} = 0$ , or equivalently,  $a_0 = 0$ .

The kernel consists of all polynomials of the form  $a_1x$ .

Since  $\ker(J) \neq \{0\}$ , by Theorem 8.2.1,  $J$  is not one-to-one.

40. For every polynomial  $q(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-2}x^{n-2} + a_{n-1}x^{n-1}$  in  $P_{n-1}$  we can construct a polynomial  $p(x) = a_0x + \frac{a_1}{2}x^2 + \frac{a_2}{3}x^3 + \cdots + \frac{a_{n-2}}{n-1}x^{n-1} + \frac{a_{n-1}}{n}x^n$  in  $P_n$  for which  $D(p(x)) = p'(x) = q(x)$ . We conclude that  $D$  is onto.

41. (a) By parts (g) and (j) of Theorem 8.2.4, the range of  $T$  cannot be  $R^n$  - it must be a proper subset of  $R^n$  instead.

For instance  $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  is the standard matrix of the orthogonal projection onto the  $x$ -axis; the

range of this transformation is the  $x$ -axis - a proper subset of  $R^2$ .

- (b) By parts (g) and (b) of Theorem 8.2.4, the kernel of  $T$  must contain at least one nonzero vector  $\mathbf{v}$ . Consequently  $T$  maps infinitely many vectors (e.g., scalar multiples  $k\mathbf{v}$ ) into  $\mathbf{0}$ .
42. (a) By parts (g) and (j) of Theorem 8.2.4, the range of  $T$  is  $R^n$ . For instance  $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$  is the standard matrix of the reflection about the  $y$ -axis; the range of this transformation is  $R^2$ .
- (b) By parts (g) and (b) of Theorem 8.2.4, the kernel of  $T$  must contain only the zero vector. Consequently  $T$  maps only  $\mathbf{0}$  into  $\mathbf{0}$ .
43. (a) Yes. If  $T_1 : R^n \rightarrow R^m$  and  $T_2 : R^m \rightarrow R^k$  are both one-to-one then for any vectors  $\mathbf{u}$  and  $\mathbf{v}$  in  $R^n$ ,  $T_2(T_1(\mathbf{u})) = T_2(T_1(\mathbf{v}))$  must imply  $T_1(\mathbf{u}) = T_1(\mathbf{v})$  (since  $T_2$  is one-to-one), which further implies that  $\mathbf{u} = \mathbf{v}$  (since  $T_1$  is one-to-one), therefore the composition  $T_2 \circ T_1 : R^n \rightarrow R^k$  is also one-to-one.
- (b) Yes. For instance,  $T_1(x_1, x_2) = (x_1, x_2, 0)$  is one-to-one but  $T_2(x_1, x_2, x_3) = (x_1, x_2)$  is not. However, the composition  $T_2(T_1(x_1, x_2)) = T_2(x_1, x_2, 0) = (x_1, x_2)$  is obviously one-to-one. However, if  $T_1$  is not one-to-one, then the composition  $T_2 \circ T_1$  is not one-to-one since there must exist two vectors  $\mathbf{u} \neq \mathbf{v}$  such that  $T_1(\mathbf{u}) = T_1(\mathbf{v})$  leading to  $T_2(T_1(\mathbf{u})) = T_2(T_1(\mathbf{v}))$ .
44. For any linear transformation  $T : V \rightarrow W$ ,  $\dim(V) = \text{nullity}(T) + \text{rank}(T) \geq \text{rank}(T)$ .  
If  $T$  is onto then  $\text{rank}(T) = \dim(W)$ . Consequently,  $\dim(V) \geq \dim(W)$ .

### True-False Exercises

- (a) True. This is equivalent to Definition 1.
- (b) False. If  $T(\mathbf{u}) = T(\mathbf{v})$ , then  $\mathbf{v}$  is the unique vector in  $V$  with image  $T(\mathbf{v})$ . Therefore  $\mathbf{u} = \mathbf{v}$ .
- (c) True.
- (d) True. For  $T$  to have an inverse, it must be one-to-one.
- (e) False.  $T^{-1}$  does not exist.
- (f) True. If  $T_1$  is not one-to-one then there is some nonzero vector  $\mathbf{v}_1$  with  $T_1(\mathbf{v}_1) = \mathbf{0}$ . Thus  $(T_2 \circ T_1)(\mathbf{v}_1) = T_2(\mathbf{0}) = \mathbf{0}$  and  $\ker(T_2 \circ T_1) \neq \{\mathbf{0}\}$ .
- (g) True. This follows from parts (b) and (j) of Theorem 8.2.4.
- (h) False. For instance,  $T_A$  and  $T_B$  where  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$  are two matrix operators on  $R^2$  whose composition is not commutative.
- (i) True. This is stated following the proof of Theorem 8.2.3.
- (j) True. This follows from parts (t) and (v) of Theorem 8.2.4.

## 8.3 Isomorphism

1. The transformation is an isomorphism.
2. The transformation is not an isomorphism(not onto).
3. The transformation is an isomorphism.
4. The transformation is not an isomorphism(not a linear transformation).
5. The transformation is not an isomorphism (not a linear transformation).
6. The transformation is an isomorphism.
7. The transformation is an isomorphism.
8. The transformation is not an isomorphism(not onto).

$$9. \quad (a) \quad T\left(\begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}\right) = \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix}$$

$$(b) \quad T_1\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \quad \text{and} \quad T_2\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a \\ c \\ b \\ d \end{bmatrix}.$$

10. (a) If  $p(x)$  is a polynomial and  $p(0) = 0$ , then  $p(x)$  has constant term 0.

$$T(ax^3 + bx^2 + cx) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$(b) \quad T(a + b\sin(x) + c\cos(x)) = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

11.  $\det(A) = -3 \neq 0$ , so by Theorem 8.2.4  $T_A$  is one-to-one and onto. Consequently,  $T_A$  is an isomorphism.
12.  $\det(A) = 0$ , so by Theorem 8.2.4  $T_A$  is not one-to-one. Consequently,  $T_A$  is not an isomorphism.

13. The reduced row echelon form of  $A$  is  $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ . The solution space  $W$  contains vectors

$(x_1, x_2, x_3, x_4)$  such that  $x_1 = -r - s - t$ ,  $x_2 = r$ ,  $x_3 = s$ ,  $x_4 = t$  so

$(x_1, x_2, x_3, x_4) = (-r - s - t, r, s, t) = r(-1, 1, 0, 0) + s(-1, 0, 1, 0) + t(-1, 0, 0, 1)$  hence  $\dim(W) = 3$ .

$(-r - s - t, r, s, t) \rightarrow (r, s, t)$  is an isomorphism between  $W$  and  $R^3$ .

14. The reduced row echelon form of  $A$  is  $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ . The solution space  $W$  contains vectors

$(x_1, x_2, x_3, x_4)$  such that  $x_1 = -s$ ,  $x_2 = -t$ ,  $x_3 = s$ ,  $x_4 = t$  so

$(x_1, x_2, x_3, x_4) = (-s, -t, s, t) = s(-1, 0, 1, 0) + t(0, -1, 0, 1)$  hence  $\dim(W) = 2$ .

$(-s, -t, s, t) \rightarrow (s, t)$  is an isomorphism between  $W$  and  $R^2$ .

15. Let us denote the given transformation by  $T$ .  $T$  is a linear transformation, since for any  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  and

$B = \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}$  in  $M_{22}$  and for any scalar  $k$ , we have

$$T(kA) = T\left(\begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}\right) = \begin{bmatrix} ka \\ ka + kb \\ ka + kb + kc \\ ka + kb + kc + kd \end{bmatrix} = k \begin{bmatrix} a \\ a + b \\ a + b + c \\ a + b + c + d \end{bmatrix} = kT(A) \text{ and}$$

$$T(A+B) = T\left(\begin{bmatrix} a+a' & b+b' \\ c+c' & d+d' \end{bmatrix}\right) = \begin{bmatrix} a+a' \\ a+a'+b+b' \\ a+a'+b+b'+c+c' \\ a+a'+b+b'+c+c'+d+d' \end{bmatrix}$$

$$= \begin{bmatrix} a \\ a+b \\ a+b+c \\ a+b+c+d \end{bmatrix} + \begin{bmatrix} a' \\ a'+b' \\ a'+b'+c' \\ a'+b'+c'+d' \end{bmatrix} = T(A) + T(B).$$

$$\text{By inspection, } \begin{bmatrix} a \\ a+b \\ a+b+c \\ a+b+c+d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ implies } a=b=c=d=0, \text{ therefore } \ker(T) = \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\} \text{ so } T \text{ is}$$

one-to-one.

$\text{rank}(T) = \dim(M_{22}) - \text{nullity}(T) = 4 - 0 = 4$  hence  $T$  is onto.

We conclude that  $T$  is an isomorphism.

16. Let us denote the given transformation by  $T$ .

By inspection,  $T\left(\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$  therefore  $\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$  is in  $\ker(T)$  so  $T$  is not one-to-one.

We conclude that  $T$  is not an isomorphism.

17. Yes,  $(a, b) \rightarrow (0, a, b)$  is an isomorphism between  $R^2$  and the  $yz$ -plane in  $R^3$ .

18. (a)  $M_{mn}$  is isomorphic to  $R^k$  if  $k = mn$  (since that makes dimensions of both spaces equal).

(b)  $M_{mn}$  is isomorphic to  $P_k$  if  $k = mn - 1$  (since that makes dimensions of both spaces equal).

19. No. By inspection,  $T(x^2 - x) = T(0) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$  so  $T$  is not one-to-one.

20. Let us denote the given transformation by  $T$ .  $T$  is a linear transformation, since for any  $A = \begin{bmatrix} a_0 & a_1 \\ a_2 & a_3 \end{bmatrix}$  and

$B = \begin{bmatrix} b_0 & b_1 \\ b_2 & b_3 \end{bmatrix}$  in  $M_{22}$  and for any scalar  $k$ , we have

$$T(kA) = T\left(\begin{bmatrix} ka_0 & ka_1 \\ ka_2 & ka_3 \end{bmatrix}\right) = ka_0 + ka_1x + ka_2x^2 + ka_3x^3 = k(a_0 + a_1x + a_2x^2 + a_3x^3) = kT(A)$$

$$T(A+B) = T\left(\begin{bmatrix} a_0+b_0 & a_1+b_1 \\ a_2+b_2 & a_3+b_3 \end{bmatrix}\right) = (a_0+b_0) + (a_1+b_1)x + (a_2+b_2)x^2 + (a_3+b_3)x^3$$

$$= (a_0 + a_1x + a_2x^2 + a_3x^3) + (b_0 + b_1x + b_2x^2 + b_3x^3) = T(A) + T(B).$$

By inspection,  $a_0 + a_1x + a_2x^2 + a_3x^3 = 0$  implies  $a_0 = a_1 = a_2 = a_3 = 0$  so  $\ker(T) = \left\{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\right\}$ .

Therefore  $T$  is one-to-one.

$\text{rank}(T) = \dim(M_{22}) - \text{nullity}(T) = 4 - 0 = 4$  hence  $T$  is onto.

We conclude that  $T$  is an isomorphism.

Furthermore,  $T(A), T(B) = a_0b_0 + a_1b_1 + a_2b_2 + a_3b_3 = A, B$ , so  $T$  is an inner product space isomorphism.

### True-False Exercises

(a) False.  $\dim(R^2) = 2$  while  $\dim(P_2) = 3$ .

(b) True. If  $\ker(T) = \{0\}$  then  $\text{rank}(T) = 4$  so  $T$  is one-to-one and onto.

(c) False.  $\dim(M_{33}) = 9$  while  $\dim(P_9) = 10$ .

- (d) True. For instance, if  $V$  consists of all matrices of the form  $\begin{bmatrix} a & b & 0 \\ c & d & 0 \end{bmatrix}$ , then  $T\left(\begin{bmatrix} a & b & 0 \\ c & d & 0 \end{bmatrix}\right) = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$  is an isomorphism  $T: V \rightarrow R^4$ .
- (e) True.
- (f) True. For instance if  $V$  consists of all vectors in  $R^{n+1}$  of the form  $(x_1, \dots, x_n, 0)$ , then  $T(x_1, \dots, x_n) = (x_1, \dots, x_n, 0)$  is an isomorphism  $T: R^n \rightarrow V$ .

## 8.4 Matrices for General Linear Transformations

1. (a)  $T(\mathbf{u}_1) = T(1) = x = \mathbf{v}_2$ ,  $T(\mathbf{u}_2) = T(x) = x^2 = \mathbf{v}_3$ , and  $T(\mathbf{u}_3) = T(x^2) = x^3 = \mathbf{v}_4$  therefore

$$[T(\mathbf{u}_1)]_{B'} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, [T(\mathbf{u}_2)]_{B'} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \text{ and } [T(\mathbf{u}_3)]_{B'} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \text{ thus } [T]_{B',B} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) By inspection,  $[\mathbf{x}]_B = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix}$  so  $[T]_{B',B}[\mathbf{x}]_B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ c_0 \\ c_1 \\ c_2 \end{bmatrix}.$

On the other hand,  $[T(\mathbf{x})]_{B'} = [c_0x + c_1x^2 + c_2x^3]_{B'} = \begin{bmatrix} 0 \\ c_0 \\ c_1 \\ c_2 \end{bmatrix}$ . Formula (5) is satisfied.

2. (a) From the definition of  $T$  we have  $T(1) = 1$ ,  $T(x) = 1 - 2x$ , and  $T(x^2) = -3x$ . By inspection,

$$[T(1)]_{B'} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, [T(x)]_{B'} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \text{ and } [T(x^2)]_{B'} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}, \text{ therefore } [T]_{B',B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & -3 \end{bmatrix}.$$

(b) By inspection,  $[\mathbf{x}]_B = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix}$  so  $[T]_{B',B}[\mathbf{x}]_B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_0 + c_1 \\ -2c_1 - 3c_2 \end{bmatrix}$ .

On the other hand,  $T(\mathbf{x}) = (c_0 + c_1) - (2c_1 + 3c_2)x$  and, by inspection,  $[T(\mathbf{x})]_{B'} = \begin{bmatrix} c_0 + c_1 \\ -(2c_1 + 3c_2) \end{bmatrix}$ .

Formula (5) is satisfied.

3. (a)  $T(1) = 1$ ;  $T(x) = x - 1 = -1 + x$ ;  $T(x^2) = (x - 1)^2 = 1 - 2x + x^2$

Thus the matrix for  $T$  relative to  $B$  is  $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$ .

(b)  $[T]_B[\mathbf{x}]_B = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_0 - a_1 + a_2 \\ a_1 - 2a_2 \\ a_2 \end{bmatrix}$ . For  $\mathbf{x} = a_0 + a_1x + a_2x^2$ ,

$T(\mathbf{x}) = a_0 + a_1(x - 1) + a_2(x - 1)^2 = a_0 - a_1 + a_2 + (a_1 - 2a_2)x + a_2x^2$ ,

so  $[T(\mathbf{x})]_B = \begin{bmatrix} a_0 - a_1 + a_2 \\ a_1 - 2a_2 \\ a_2 \end{bmatrix}$ .

4. (a) From the definition of  $T$  we have  $T(\mathbf{u}_1) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$  and  $T(\mathbf{u}_2) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ . To find the coordinate vectors,

we solve the systems

$$\begin{array}{rcl} a_1 - a_2 & = & 0 \\ a_1 & = & 2 \end{array} \quad \text{and} \quad \begin{array}{rcl} b_1 - b_2 & = & -1 \\ b_1 & = & -1 \end{array}$$

so clearly we have  $[T(\mathbf{u}_1)]_B = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$  and  $[T(\mathbf{u}_2)]_B = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ . Therefore  $[T]_B = \begin{bmatrix} 2 & -1 \\ 2 & 0 \end{bmatrix}$ .

(b) For an arbitrary  $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$  in  $R^2$ , solving the system

$$\begin{array}{rcl} c_1 - c_2 & = & x \\ c_1 & = & y \end{array}$$

yields  $[\mathbf{x}]_B = \begin{bmatrix} y \\ y - x \end{bmatrix}$  therefore  $[T]_B[\mathbf{x}]_B = \begin{bmatrix} 2 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} y \\ y - x \end{bmatrix} = \begin{bmatrix} x + y \\ 2y \end{bmatrix}$ .

On the other hand,  $T(\mathbf{x}) = \begin{bmatrix} x - y \\ x + y \end{bmatrix}$ . Solving the system

$$\begin{array}{rcl} c_1 - c_2 & = & x - y \\ c_1 & = & x + y \end{array}$$

yields  $[T(\mathbf{x})]_B = \begin{bmatrix} x+y \\ 2y \end{bmatrix}$  showing that Formula (8) holds for every  $\mathbf{x}$  in  $R^2$ .

$$5. \quad (a) \quad T(\mathbf{u}_1) = T\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 7 \\ -1 \\ 0 \end{bmatrix} = 0\mathbf{v}_1 - \frac{1}{2}\mathbf{v}_2 + \frac{8}{3}\mathbf{v}_3; \quad T(\mathbf{u}_2) = T\left(\begin{bmatrix} -2 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ 2 \\ 0 \end{bmatrix} = 0\mathbf{v}_1 + \mathbf{v}_2 + \frac{4}{3}\mathbf{v}_3$$

$$[T]_{B', B} = \begin{bmatrix} 0 & 0 \\ -\frac{1}{2} & 1 \\ \frac{8}{3} & \frac{4}{3} \end{bmatrix}$$

(b) For an arbitrary  $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$  in  $R^2$ , solving the system

$$\begin{aligned} c_1 - 2c_2 &= x \\ 3c_1 + 4c_2 &= y \end{aligned}$$

$$\text{yields } [\mathbf{x}]_B = \begin{bmatrix} \frac{2x+y}{5} \\ \frac{-3x+y}{10} \end{bmatrix} \text{ therefore } [T]_{B', B} [\mathbf{x}]_B = \begin{bmatrix} 0 & 0 \\ -\frac{1}{2} & 1 \\ \frac{8}{3} & \frac{4}{3} \end{bmatrix} \begin{bmatrix} \frac{2x+y}{5} \\ \frac{-3x+y}{10} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{x}{2} \\ \frac{2}{3}x + \frac{2}{3}y \end{bmatrix}.$$

On the other hand,  $T(\mathbf{x}) = \begin{bmatrix} x+2y \\ -x \\ 0 \end{bmatrix}$ . Solving the system

$$\begin{aligned} c_1 + 2c_2 + 3c_3 &= x+2y \\ c_1 + 2c_2 &= -x \\ c_1 &= 0 \end{aligned}$$

$$\text{yields } [T(\mathbf{x})]_{B'} = \begin{bmatrix} 0 \\ -\frac{x}{2} \\ \frac{2}{3}x + \frac{2}{3}y \end{bmatrix} \text{ showing that Formula (5) holds for every } \mathbf{x} \text{ in } R^2.$$

$$6. \quad (a) \quad \text{We have } [T(\mathbf{v}_1)]_B = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, [T(\mathbf{v}_2)]_B = \begin{bmatrix} -\frac{3}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \text{ and } [T(\mathbf{v}_3)]_B = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}.$$

$$\text{Therefore } [T]_B = \begin{bmatrix} 1 & -\frac{3}{2} & \frac{1}{2} \\ -1 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

(b) For an arbitrary  $\mathbf{x} = (x_1, x_2, x_3)$  in  $R^3$ , solving the system

$$\begin{aligned} a_1 &+ a_3 &= x_1 \\ a_2 &+ a_3 &= x_2 \\ a_1 &+ a_2 &= x_3 \end{aligned}$$

$$\text{yields } [\mathbf{x}]_B = \begin{bmatrix} \frac{1}{2}(x_1 - x_2 + x_3) \\ \frac{1}{2}(-x_1 + x_2 + x_3) \\ \frac{1}{2}(x_1 + x_2 - x_3) \end{bmatrix} \text{ therefore}$$

$$[T]_B [\mathbf{x}]_B = \begin{bmatrix} 1 & -\frac{3}{2} & \frac{1}{2} \\ -1 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2}(x_1 - x_2 + x_3) \\ \frac{1}{2}(-x_1 + x_2 + x_3) \\ \frac{1}{2}(x_1 + x_2 - x_3) \end{bmatrix} = \begin{bmatrix} \frac{3}{2}x_1 - x_2 - \frac{1}{2}x_3 \\ -\frac{1}{2}x_1 + x_2 - \frac{1}{2}x_3 \\ -\frac{1}{2}x_1 + \frac{1}{2}x_3 \end{bmatrix}.$$

On the other hand, solving the system

$$\begin{aligned} b_1 + b_3 &= x_1 - x_2 \\ b_2 + b_3 &= x_2 - x_1 \\ b_1 + b_2 &= x_1 - x_3 \end{aligned}$$

$$\text{yields } [T(\mathbf{x})]_B = \begin{bmatrix} \frac{3}{2}x_1 - x_2 - \frac{1}{2}x_3 \\ -\frac{1}{2}x_1 + x_2 - \frac{1}{2}x_3 \\ -\frac{1}{2}x_1 + \frac{1}{2}x_3 \end{bmatrix} \text{ showing that Formula (8) holds for every } \mathbf{x} \text{ in } R^3.$$

(c) The augmented matrix of the homogeneous system

$$\begin{aligned} x_1 - x_2 &= 0 \\ -x_1 + x_2 &= 0 \\ x_1 - x_3 &= 0 \end{aligned}$$

$$\text{has the reduced row echelon form } \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \text{ Since the system has nontrivial solutions}$$

(e.g.,  $(1,1,1)$ )  $\ker(T) \neq \{(0,0,0)\}$  so that  $T$  is not one-to-one.

7. (a) We have  $T(1) = 1$ ,  $T(x) = 2x + 1 = 1 + 2x$ , and  $T(x^2) = (2x + 1)^2 = 1 + 4x + 4x^2$ .

$$\text{Therefore, } [T]_B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 4 \end{bmatrix}.$$

(b) Step 1. The coordinate vector of  $\mathbf{x} = 2 - 3x + 4x^2$  with respect to the basis  $B$  is

$$[\mathbf{x}]_B = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

$$\text{Step 2. } [T]_B [\mathbf{x}]_B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 10 \\ 16 \end{bmatrix} = [T(\mathbf{x})]_B.$$

Step 3. Reconstructing  $T(\mathbf{x})$  from the coordinate vector obtained in Step 2, we obtain

$$T(\mathbf{x}) = 3(1) + 10(x) + 16(x^2) = 3 + 10x + 16x^2.$$

$$(c) \quad T(2 - 3x + 4x^2) = 2 - 3(2x + 1) + 4(2x + 1)^2 = 2 - 6x - 3 + 16x^2 + 16x + 4 = 3 + 10x + 16x^2$$

8. (a) From the definition of  $T$  we have  $T(1) = x$ ,  $T(x) = x^2 - 3x$ , and

$$T(x^2) = x(x - 3)^2 = x^3 - 6x^2 + 9x.$$

The corresponding coordinate vectors are  $[T(1)]_{B'} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ ,  $[T(x)]_{B'} = \begin{bmatrix} 0 \\ -3 \\ 1 \\ 0 \end{bmatrix}$ , and

$$[T(x^2)]_{B'} = \begin{bmatrix} 0 \\ 9 \\ -6 \\ 1 \end{bmatrix}, \text{ therefore } [T]_{B', B} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & -3 & 9 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) Step 1. The coordinate vector of  $\mathbf{x} = 1 + x - x^2$  with respect to the basis  $B$  is  $[\mathbf{x}]_B = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ .

$$\text{Step 2. } [T]_{B', B} [\mathbf{x}]_B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & -3 & 9 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -11 \\ 7 \\ -1 \end{bmatrix} = [T(\mathbf{x})]_{B'}.$$

Step 3. Reconstructing  $T(\mathbf{x})$  from the coordinate vector obtained in Step 2, we obtain

$$T(\mathbf{x}) = 0(1) - 11(x) + 7(x^2) - 1(x^3) = -11x + 7x^2 - x^3.$$

$$(c) \quad T(1 + x - x^2) = x(1 + (x - 3) - (x - 3)^2) = -11x + 7x^2 - x^3.$$

9. (a) Since  $A$  is the matrix for  $T$  relative to  $B$ ,  $A = \begin{bmatrix} [T(\mathbf{v}_1)]_B & [T(\mathbf{v}_2)]_B \end{bmatrix}$ .

$$\text{That is, } [T(\mathbf{v}_1)]_B = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \text{ and } [T(\mathbf{v}_2)]_B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

$$(b) \quad \text{Since } [T(\mathbf{v}_1)]_B = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad T(\mathbf{v}_1) = 1\mathbf{v}_1 - 2\mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \begin{bmatrix} -2 \\ 8 \end{bmatrix} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}.$$

$$\text{Similarly, } T(\mathbf{v}_2) = 3\mathbf{v}_1 + 5\mathbf{v}_2 = \begin{bmatrix} 3 \\ 9 \end{bmatrix} + \begin{bmatrix} -5 \\ 20 \end{bmatrix} = \begin{bmatrix} -2 \\ 29 \end{bmatrix}.$$

(c) If  $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ , then

$$x_1 = c_1 - c_2$$

$$x_2 = 3c_1 + 4c_2$$

Solving for  $c_1$  and  $c_2$  gives  $c_1 = \frac{4}{7}x_1 + \frac{1}{7}x_2$ ,  $c_2 = -\frac{3}{7}x_1 + \frac{1}{7}x_2$ , so

$$\begin{aligned} T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) &= c_1 T(\mathbf{v}_1) + c_2 T(\mathbf{v}_2) = \left(\frac{4}{7}x_1 + \frac{1}{7}x_2\right) \begin{bmatrix} 3 \\ -5 \end{bmatrix} + \left(-\frac{3}{7}x_1 + \frac{1}{7}x_2\right) \begin{bmatrix} -2 \\ 29 \end{bmatrix} = \begin{bmatrix} \frac{18}{7}x_1 + \frac{1}{7}x_2 \\ -\frac{107}{7}x_1 + \frac{24}{7}x_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{18}{7} & \frac{1}{7} \\ -\frac{107}{7} & \frac{24}{7} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \end{aligned}$$

(d)  $T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} \frac{18}{7} & \frac{1}{7} \\ -\frac{107}{7} & \frac{24}{7} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{19}{7} \\ -\frac{83}{7} \end{bmatrix}$

10. (a) By Formula (4),  $[T(\mathbf{v}_1)]_{B'} = \begin{bmatrix} 3 \\ 1 \\ -3 \end{bmatrix}$ ,  $[T(\mathbf{v}_2)]_{B'} = \begin{bmatrix} -2 \\ 6 \\ 0 \end{bmatrix}$ ,  $[T(\mathbf{v}_3)]_{B'} = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$ , and  $[T(\mathbf{v}_4)]_{B'} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ .

(b) Reconstructing the vectors using their coordinate vectors obtained in part (a) yields

$$\begin{aligned} T(\mathbf{v}_1) &= 3 \begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix} + 1 \begin{bmatrix} -7 \\ 8 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} -6 \\ 9 \\ 1 \end{bmatrix} = \begin{bmatrix} 11 \\ 5 \\ 22 \end{bmatrix} & T(\mathbf{v}_2) &= -2 \begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix} + 6 \begin{bmatrix} -7 \\ 8 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} -6 \\ 9 \\ 1 \end{bmatrix} = \begin{bmatrix} -42 \\ 32 \\ -10 \end{bmatrix} \\ T(\mathbf{v}_3) &= 1 \begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix} + 2 \begin{bmatrix} -7 \\ 8 \\ 1 \end{bmatrix} + 7 \begin{bmatrix} -6 \\ 9 \\ 1 \end{bmatrix} = \begin{bmatrix} -56 \\ 87 \\ 17 \end{bmatrix} & T(\mathbf{v}_4) &= 0 \begin{bmatrix} 0 \\ 8 \\ 8 \end{bmatrix} + 1 \begin{bmatrix} -7 \\ 8 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} -6 \\ 9 \\ 1 \end{bmatrix} = \begin{bmatrix} -13 \\ 17 \\ 2 \end{bmatrix} \end{aligned}$$

(c) We follow the procedure of Example 10 in Section 8.1. First of all, we express  $\mathbf{x} = (x_1, x_2, x_3, x_4)$  as a linear combination of the vectors in  $B$ . The corresponding linear system

$$\begin{aligned} 2c_2 + c_3 + 6c_4 &= x_1 \\ c_1 + c_2 + 4c_3 + 9c_4 &= x_2 \\ c_1 - c_2 - c_3 + 4c_4 &= x_3 \\ c_1 - c_2 + 2c_3 + 2c_4 &= x_4 \end{aligned}$$

has the augmented matrix whose reduced row echelon form is

$$\left[ \begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{9}{2}x_1 - 4x_2 - \frac{1}{2}x_3 + \frac{11}{2}x_4 \\ 0 & 1 & 0 & 0 & \frac{5}{2}x_1 - 2x_2 - \frac{1}{2}x_3 + \frac{5}{2}x_4 \\ 0 & 0 & 1 & 0 & -\frac{2}{5}x_1 + \frac{2}{5}x_2 - \frac{1}{5}x_3 - \frac{1}{5}x_4 \\ 0 & 0 & 0 & 1 & -\frac{3}{5}x_1 + \frac{3}{5}x_2 + \frac{1}{5}x_3 - \frac{4}{5}x_4 \end{array} \right].$$

This yields  $c_1 = \frac{9}{2}x_1 - 4x_2 - \frac{1}{2}x_3 + \frac{11}{2}x_4$ ,  $c_2 = \frac{5}{2}x_1 - 2x_2 - \frac{1}{2}x_3 + \frac{5}{2}x_4$ ,

$c_3 = -\frac{2}{5}x_1 + \frac{2}{5}x_2 - \frac{1}{5}x_3 - \frac{1}{5}x_4$ ,  $c_4 = -\frac{3}{5}x_1 + \frac{3}{5}x_2 + \frac{1}{5}x_3 - \frac{4}{5}x_4$ . Therefore

$$\begin{aligned} T(x_1, x_2, x_3, x_4) &= \left(\frac{9}{2}x_1 - 4x_2 - \frac{1}{2}x_3 + \frac{11}{2}x_4\right) \begin{bmatrix} 11 \\ 5 \\ 22 \end{bmatrix} + \left(\frac{5}{2}x_1 - 2x_2 - \frac{1}{2}x_3 + \frac{5}{2}x_4\right) \begin{bmatrix} -42 \\ 32 \\ -10 \end{bmatrix} \\ &+ \left(-\frac{2}{5}x_1 + \frac{2}{5}x_2 - \frac{1}{5}x_3 - \frac{1}{5}x_4\right) \begin{bmatrix} -56 \\ 87 \\ 17 \end{bmatrix} + \left(-\frac{3}{5}x_1 + \frac{3}{5}x_2 + \frac{1}{5}x_3 - \frac{4}{5}x_4\right) \begin{bmatrix} -13 \\ 17 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{253}{10}x_1 + \frac{49}{5}x_2 + \frac{241}{10}x_3 - \frac{229}{10}x_4 \\ \frac{115}{2}x_1 - 39x_2 - \frac{65}{2}x_3 + \frac{153}{2}x_4 \\ 66x_1 - 60x_2 - 9x_3 + 91x_4 \end{bmatrix}. \end{aligned}$$

$$(d) \quad T \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \end{pmatrix} = \begin{bmatrix} -31 \\ 37 \\ 12 \end{bmatrix}$$

11. (a) Since  $A$  is the matrix for  $T$  relative to  $B$ , the columns of  $A$  are  $[T(\mathbf{v}_1)]_B$ ,  $[T(\mathbf{v}_2)]_B$ , and  $[T(\mathbf{v}_3)]_B$ ,

respectively. That is,  $[T(\mathbf{v}_1)]_B = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}$ ,  $[T(\mathbf{v}_2)]_B = \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}$ , and  $[T(\mathbf{v}_3)]_B = \begin{bmatrix} -1 \\ 5 \\ 4 \end{bmatrix}$ .

$$(b) \quad \text{Since } [T(\mathbf{v}_1)]_B = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix},$$

$$T(\mathbf{v}_1) = \mathbf{v}_1 + 2\mathbf{v}_2 + 6\mathbf{v}_3 = 3x + 3x^2 - 2 + 6x + 4x^2 + 18 + 42x + 12x^2 = 16 + 51x + 19x^2.$$

$$\text{Similarly, } T(\mathbf{v}_2) = 3\mathbf{v}_1 - 2\mathbf{v}_3 = 9x + 9x^2 - 6 - 14x - 4x^2 = -6 - 5x + 5x^2, \text{ and}$$

$$\begin{aligned} T(\mathbf{v}_3) &= -\mathbf{v}_1 + 5\mathbf{v}_2 + 4\mathbf{v}_3 = -3x - 3x^2 - 5 + 15x + 10x^2 + 12 + 28x + 8x^2 \\ &= 7 + 40x + 15x^2. \end{aligned}$$

(c) If  $a_0 + a_1x + a_2x^2 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$

$$= c_1(3x + 3x^2) + c_2(-1 + 3x + 2x^2) + c_3(3 + 7x + 2x^2) \text{ then}$$

$$a_0 = -c_2 + 3c_3, \quad a_1 = 3c_1 + 3c_2 + 7c_3, \quad a_2 = 3c_1 + 2c_2 + 2c_3.$$

Solving for  $c_1$ ,  $c_2$ , and  $c_3$  gives

$$c_1 = \frac{1}{3}(a_0 - a_1 + 2a_2), \quad c_2 = \frac{1}{8}(-5a_0 + 3a_1 - 3a_2), \quad c_3 = \frac{1}{8}(a_0 + a_1 - a_2), \text{ so}$$

$$T(a_0 + a_1x + a_2x^2) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + c_3T(\mathbf{v}_3)$$

$$= \frac{1}{3}(a_0 - a_1 + 2a_2)(16 + 51x + 19x^2) + \frac{1}{8}(-5a_0 + 3a_1 - 3a_2)(-6 - 5x + 5x^2)$$

$$\begin{aligned}
& +\frac{1}{8}(a_0 + a_1 - a_2)(7 + 40x + 15x^2) \\
& = \frac{239a_0 - 161a_1 + 289a_2}{24} + \frac{201a_0 - 111a_1 + 247a_2}{8}x + \frac{61a_0 - 31a_1 + 107a_2}{12}x^2.
\end{aligned}$$

(d) In  $1 + x^2$ ,  $a_0 = 1$ ,  $a_1 = 0$ ,  $a_2 = 1$ .

$$T(1 + x^2) = \frac{239+289}{24} + \frac{201+247}{8}x + \frac{61+107}{12}x^2 = 22 + 56x + 14x^2$$

12. (a)  $(T_2 \circ T_1)(1) = T_2(T_1(1)) = T_2(x) = 2x + 1$ ;

$$(T_2 \circ T_1)(x) = T_2(T_1(x)) = T_2(x^2) = (2x + 1)^2 = 4x^2 + 4x + 1;$$

$$[(T_2 \circ T_1)(1)]_{B'} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \text{ and } [(T_2 \circ T_1)(x)]_{B'} = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} \text{ therefore } [T_2 \circ T_1]_{B', B} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 0 & 4 \end{bmatrix};$$

$$T_2(1) = 1; T_2(x) = 2x + 1; T_2(x^2) = (2x + 1)^2 = 4x^2 + 4x + 1;$$

$$[T_2(1)]_{B'} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, [T_2(x)]_{B'} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \text{ and } [T_2(x^2)]_{B'} = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} \text{ therefore } [T_2]_{B'} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 4 \end{bmatrix};$$

$$T_1(1) = x; T_1(x) = x^2; [T_1(1)]_{B'} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ and } [T_1(x)]_{B'} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ therefore } [T_1]_{B', B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

(b) By Theorem 8.4.1,  $[T_2 \circ T_1]_{B', B} = [T_2]_{B'} [T_1]_{B', B}$ .

$$(c) [T_2]_{B'} [T_1]_{B', B} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 0 & 4 \end{bmatrix} = [T_2 \circ T_1]_{B', B}$$

13. (a)  $(T_2 \circ T_1)(1) = T_2(2) = 6x$  and  $(T_2 \circ T_1)(x) = T_2(-3x) = -9x^2$  so  $[T_2 \circ T_1]_{B', B} = \begin{bmatrix} 0 & 0 \\ 6 & 0 \\ 0 & -9 \\ 0 & 0 \end{bmatrix};$

$$T_1(1) = 2 \text{ and } T_1(x) = -3x \text{ so } [T_1]_{B', B} = \begin{bmatrix} 2 & 0 \\ 0 & -3 \\ 0 & 0 \end{bmatrix};$$

$$T_2(1) = 3x, T_2(x) = 3x^2, \text{ and } T_2(x^2) = 3x^3 \text{ so } [T_2]_{B', B'} = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

(b)  $[T_2 \circ T_1]_{B', B} = [T_2]_{B', B'} [T_1]_{B', B}$

$$(c) \quad \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 6 & 0 \\ 0 & -9 \\ 0 & 0 \end{bmatrix}$$

$$14. \quad [T(\mathbf{v}_1)]_B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, [T(\mathbf{v}_2)]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, [T(\mathbf{v}_3)]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \text{ and } [T(\mathbf{v}_4)]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ therefore } [T]_B = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$15. \quad (a) \quad \text{Since } T(1) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, T(x) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, T(x^2) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \text{ we have } [T(1)]_B = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

$$[T(x)]_B = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \text{ and } [T(x^2)]_B = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}. \text{ Consequently, } [T]_{B,B'} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

$$\text{Since } T(1) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, T(1+x) = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \text{ and } T(1+x^2) = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \text{ we have } [T(1)]_B = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

$$[T(1+x)]_B = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \text{ and } [T(1+x^2)]_B = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}. \text{ Consequently, } [T]_{B,B'} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 0 & 2 \\ 1 & 1 & 1 \end{bmatrix}.$$

(b) Applying the three-step procedure to the bases  $B'$  and  $B$  we obtain:

$$\text{Step 1.} \quad \text{The coordinate vector of } \mathbf{x} = 2 + 2x + x^2 \text{ with respect to the basis } B' \text{ is } [\mathbf{x}]_{B'} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}.$$

$$\text{Step 2.} \quad [T]_{B,B'} [\mathbf{x}]_{B'} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 1 \\ 2 \end{bmatrix} = [T(\mathbf{x})]_B.$$

Step 3. Reconstructing  $T(\mathbf{x})$  from the coordinate vector obtained in Step 2, we obtain

$$T(\mathbf{x}) = 2 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 5 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}.$$

Applying the three-step procedure to the bases  $B''$  and  $B$  we obtain:

Step 1. The coordinate vector of  $\mathbf{x} = 2 + 2x + x^2$  with respect to the basis  $B''$  is  $[\mathbf{x}]_{B''} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ .

To find the coordinate vector of  $\mathbf{x} = 2 + 2x + x^2$  with respect to the basis  $B'$  we solve the system

$$\begin{aligned} c_1 + c_2 + c_3 &= 2 \\ c_2 &= 2 \\ &+ c_3 = 1 \end{aligned}$$

Back-substitution yields  $[\mathbf{x}]_{B''} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ .

Step 2.  $[T]_{B, B''} [\mathbf{x}]_{B''} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 0 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 1 \\ 2 \end{bmatrix} = [T(\mathbf{x})]_B.$

Step 3. Reconstructing  $T(\mathbf{x})$  from the coordinate vector obtained in Step 2, we obtain

$$T(\mathbf{x}) = 2 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 5 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}.$$

(c)  $T(2 + 2x + x^2) = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}$

16. (a) Since  $T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = T\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$  and  $T\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ , we have

$$\left[T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right)\right]_{B'} = \left[T\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right)\right]_{B'} = \left[T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right)\right]_{B'} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \left[T\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right)\right]_{B'} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ and}$$

$$\left[T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right)\right]_{B'} = \left[T\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right)\right]_{B'} = \left[T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right)\right]_{B'} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \text{ and } \left[T\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right)\right]_{B'} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ therefore}$$

$$[T]_{B', B} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ and } [T]_{B'', B} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix}.$$

(b) Applying the three-step procedure to the bases  $B$  and  $B'$  we obtain:

Step 1. The coordinate vector of  $\mathbf{x} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  with respect to the basis  $B$  is  $[\mathbf{x}]_B = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ .

$$\text{Step 2.} \quad [T]_{B',B}[\mathbf{x}]_B = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = [T(\mathbf{x})]_{B'}.$$

Step 3. Reconstructing  $T(\mathbf{x})$  from the coordinate vector obtained in Step 2, we obtain

$$T(\mathbf{x}) = 6 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

Applying the three-step procedure to the bases  $B$  and  $B''$  we obtain:

$$\text{Step 1.} \quad \text{The coordinate vector of } \mathbf{x} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ with respect to the basis } B \text{ is } [\mathbf{x}]_B = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}.$$

$$\text{Step 2.} \quad [T]_{B'',B}[\mathbf{x}]_B = \begin{bmatrix} 0 & 0 & 0 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} = [T(\mathbf{x})]_{B''}.$$

Step 3. Reconstructing  $T(\mathbf{x})$  from the coordinate vector obtained in Step 2, we obtain

$$T(\mathbf{x}) = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

$$(c) \quad T\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$17. (a) \quad D(\mathbf{p}_1) = 0; D(\mathbf{p}_2) = 1; D(\mathbf{p}_3) = 2x; [D(\mathbf{p}_1)]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, [D(\mathbf{p}_2)]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \text{ and } [D(\mathbf{p}_3)]_B = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$$

$$\text{therefore } [D]_B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$(b) \quad \text{Denoting } \mathbf{p} = 6 - 6x + 24x^2, \text{ we have } [\mathbf{p}]_B = \begin{bmatrix} 6 \\ -6 \\ 24 \end{bmatrix};$$

$$[D]_B[\mathbf{p}]_B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ -6 \\ 24 \end{bmatrix} = \begin{bmatrix} -6 \\ 48 \\ 0 \end{bmatrix} = [D(\mathbf{p})]_B \text{ thus } D(\mathbf{p}) = -6\mathbf{p}_1 + 48\mathbf{p}_2 + 0\mathbf{p}_3 = -6 + 48x.$$

18. (a)  $D(\mathbf{p}_1) = 0$ ;  $D(\mathbf{p}_2) = -3$ ;  $D(\mathbf{p}_3) = -3 + 16x$ ; by inspection we obtain  $[D(\mathbf{p}_1)]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ ;

to determine the coordinate vectors  $[D(\mathbf{p}_2)]_B$  and  $[D(\mathbf{p}_3)]_B$  we solve the systems

$$\begin{array}{rclcl} 2a_1 + 2a_2 + 2a_3 & = & -3 & & 2b_1 + 2b_2 + 2b_3 = -3 \\ & - & 3a_2 & - & 3a_3 = 0 & \text{and} & - & 3b_2 & - & 3b_3 = 16 \\ & & 8a_3 & = & 0 & & & 8b_3 & = & 0 \end{array}$$

obtaining  $[D(\mathbf{p}_2)]_B = \begin{bmatrix} -\frac{3}{2} \\ 0 \\ 0 \end{bmatrix}$  and  $[D(\mathbf{p}_3)]_B = \begin{bmatrix} \frac{23}{6} \\ -\frac{16}{3} \\ 0 \end{bmatrix}$ . Therefore  $[D]_B = \begin{bmatrix} 0 & -\frac{3}{2} & \frac{23}{6} \\ 0 & 0 & -\frac{16}{3} \\ 0 & 0 & 0 \end{bmatrix}$ .

(b) Solving the system

$$\begin{array}{rcl} 2a_1 + 2a_2 + 2a_3 & = & 6 \\ & - & 3a_2 - 3a_3 = -6 \\ & & 8a_3 = 24 \end{array}$$

we obtain  $a_1 = 1$ ,  $a_2 = -1$ ,  $a_3 = 3$  so that  $[\mathbf{p}]_B = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ ;

$$[D]_B [\mathbf{p}]_B = \begin{bmatrix} 0 & -\frac{3}{2} & \frac{23}{6} \\ 0 & 0 & -\frac{16}{3} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 13 \\ -16 \\ 0 \end{bmatrix} = [D(\mathbf{p})]_B$$

therefore  $D(\mathbf{p}) = 13\mathbf{p}_1 - 16\mathbf{p}_2 + 0\mathbf{p}_3 = 13(2) - 16(2 - 3x) + 0(2 - 3x + 8x^2) = -6 + 48x$

19. (a)  $D(\mathbf{f}_1) = D(1) = 0$ ;  $D(\mathbf{f}_2) = D(\sin x) = \cos x$ ;  $D(\mathbf{f}_3) = D(\cos x) = -\sin x$

The matrix for  $D$  relative to this basis is  $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$ .

(b) Since  $[2 + 3\sin x - 4\cos x]_B = \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix}$ ,  $[D(2 + 3\sin x - 4\cos x)]_B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 3 \end{bmatrix}$ .

Consequently,  $D(2 + 3\sin x - 4\cos x) = (0)(1) + 4\sin x + 3\cos x = 4\sin x + 3\cos x$ .

20.  $\mathbf{x}$  is in  $V$ ;  $[\mathbf{x}]_B$  is in  $R^4$ ;  $[T(\mathbf{x})]_{B'}$  is in  $R^7$ ;  $T(\mathbf{x})$  is in  $W$

21. (a)  $[T_2 \circ T_1]_{B', B} = [T_2]_{B', B''} [T_1]_{B'', B}$

(b)  $[T_3 \circ T_2 \circ T_1]_{B', B} = [T_3]_{B', B'''} [T_2]_{B'', B''} [T_1]_{B'', B}$

22. Applying  $T$  to each basis vector for  $V$  yields  $\mathbf{0}$  (the zero vector in  $W$ ). Furthermore, the coordinate vector of  $\mathbf{0}$  with respect to any basis for  $W$  must be the zero vector in  $R^n$  where  $n = \dim(W)$ . It follows from formula (4) that the matrix for  $T$  relative to the two bases is a zero matrix.
23. The matrix for  $T$  relative to  $B$  is the matrix whose columns are the transforms of the basis vectors in  $B$  in terms of the standard basis. Since  $B$  is the standard basis for  $R^n$ , this matrix is the standard matrix for  $T$ . Also, since  $B'$  is the standard basis for  $R^m$ , the resulting transformation will give vector components relative to the standard basis.

### True-False Exercises

- (a) False. The conclusion would only be true if  $T : V \rightarrow V$  were a linear operator, i.e., if  $V = W$ .
- (b) False.
- (c) True. Since the matrix for  $T$  is invertible, by Theorem 8.4.2  $\ker(T) = \{\mathbf{0}\}$ .
- (d) False. It follows from Theorem 8.4.1 that the matrix of  $S \circ T$  relative to  $B$  is  $[S]_B[T]_B$ .
- (e) True. This follows from Theorem 8.4.2.

## 8.5 Similarity

1. (a)  $\det(A) = -2$  does not equal  $\det(B) = -1$

(b)  $\operatorname{tr}(A) = 3$  does not equal  $\operatorname{tr}(B) = -2$

2. (a)  $\det(A) = -1$  does not equal  $\det(B) = 0$

(b)  $\operatorname{tr}(A) = 2$  does not equal  $\operatorname{tr}(B) = 0$

3. Since  $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}^{-1} = \frac{1}{(3)(1)-(2)(1)} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix},$

$$[T]_{B'} = P_{B \rightarrow B'} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 6 & -10 \\ 2 & -3 \end{bmatrix}$$

4. Since  $\begin{bmatrix} 4 & 5 \\ 1 & -1 \end{bmatrix}^{-1} = \frac{1}{(4)(-1)-(5)(1)} \begin{bmatrix} -1 & -5 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{9} & \frac{5}{9} \\ \frac{1}{9} & -\frac{4}{9} \end{bmatrix},$

$$[T]_{B'} = P_{B \rightarrow B'} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} \frac{1}{9} & \frac{5}{9} \\ \frac{1}{9} & -\frac{4}{9} \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{9} & -\frac{17}{9} \\ \frac{26}{9} & \frac{37}{9} \end{bmatrix}$$

5. Since  $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}^{-1} = \frac{1}{(3)(1)-(2)(1)} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix},$

$$[T]_B = P_{B' \rightarrow B} [T]_{B'} P_{B \rightarrow B'} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 6 & 5 \end{bmatrix}$$

6. Since  $\begin{bmatrix} 4 & 5 \\ 1 & -1 \end{bmatrix}^{-1} = \frac{1}{(4)(-1)-(5)(1)} \begin{bmatrix} -1 & -5 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{9} & \frac{5}{9} \\ \frac{1}{9} & -\frac{4}{9} \end{bmatrix},$

$$[T]_B = P_{B' \rightarrow B} [T]_{B'} P_{B \rightarrow B'} = \begin{bmatrix} 4 & 5 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{9} & \frac{5}{9} \\ \frac{1}{9} & -\frac{4}{9} \end{bmatrix} = \begin{bmatrix} \frac{20}{9} & -\frac{17}{9} \\ \frac{5}{9} & \frac{16}{9} \end{bmatrix}$$

7. From the definition of  $T$  we have  $T(\mathbf{u}_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = [T(\mathbf{u}_1)]_B$  and  $T(\mathbf{u}_2) = \begin{bmatrix} -2 \\ -1 \end{bmatrix} = [T(\mathbf{u}_2)]_B$  therefore

$$[T]_B = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}. \text{ Since } [\mathbf{v}_1]_B = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \text{ and } [\mathbf{v}_2]_B = \begin{bmatrix} 7 \\ 2 \end{bmatrix}, \text{ we have } P_{B' \rightarrow B} = \begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}.$$

The inverse of  $P_{B' \rightarrow B}$  is  $P_{B' \rightarrow B}^{-1} = \begin{bmatrix} 2 & -7 \\ -1 & 4 \end{bmatrix}$ . Using Theorem 8.5.2, we obtain

$$[T]_{B'} = P_{B' \rightarrow B}^{-1} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} 2 & -7 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 11 & 20 \\ -6 & -11 \end{bmatrix}.$$

8. From the definition of  $T$  we have  $T(\mathbf{u}_1) = \begin{bmatrix} 16 \\ -2 \end{bmatrix}$  and  $T(\mathbf{u}_2) = \begin{bmatrix} -3 \\ 16 \end{bmatrix}$ . Solving the linear systems

$$\begin{array}{rcl} 2a_1 + 4a_2 & = & 16 \\ 2a_1 - a_2 & = & -2 \end{array} \quad \text{and} \quad \begin{array}{rcl} 2b_1 + 4b_2 & = & -3 \\ 2b_1 - b_2 & = & 16 \end{array}$$

yields the coordinate vectors  $[T(\mathbf{u}_1)]_B = \begin{bmatrix} \frac{4}{5} \\ \frac{18}{5} \end{bmatrix}$  and  $[T(\mathbf{u}_2)]_B = \begin{bmatrix} \frac{61}{10} \\ -\frac{19}{5} \end{bmatrix}$  therefore  $[T]_B = \begin{bmatrix} \frac{4}{5} & \frac{61}{10} \\ \frac{18}{5} & -\frac{19}{5} \end{bmatrix}.$

From Theorem 8.5.1,  $P_{B' \rightarrow B} = [I]_{B, B'}$ . Solving the linear systems

$$\begin{array}{rcl} 2a_1 + 4a_2 & = & 18 \\ 2a_1 - a_2 & = & 8 \end{array} \quad \text{and} \quad \begin{array}{rcl} 2b_1 + 4b_2 & = & 10 \\ 2b_1 - b_2 & = & 15 \end{array}$$

yields the coordinate vectors  $[\mathbf{v}_1]_B = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$  and  $[\mathbf{v}_2]_B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$  therefore  $P_{B' \rightarrow B} = \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix}.$

The inverse of  $P_{B' \rightarrow B}$  is  $P_{B' \rightarrow B}^{-1} = \begin{bmatrix} -1 & 3 \\ 2 & -5 \end{bmatrix}$ . Using Theorem 8.5.2, we obtain

$$[T]_{B'} = P_{B' \rightarrow B}^{-1} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} -1 & 3 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} \frac{4}{5} & \frac{61}{10} \\ \frac{18}{5} & -\frac{19}{5} \end{bmatrix} \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 15 & \frac{25}{2} \\ -\frac{98}{2} & -18 \end{bmatrix}.$$

9. Denoting  $\mathbf{e}_1 = (1, 0, 0)$ ,  $\mathbf{e}_2 = (0, 1, 0)$ , and  $\mathbf{e}_3 = (0, 0, 1)$ , we have  $T(\mathbf{e}_1) = (-2, 1, 0) = (T(\mathbf{e}_1))_B$ ,

$$T(\mathbf{e}_2) = (-1, 0, 1) = (T(\mathbf{e}_2))_B, \text{ and } T(\mathbf{e}_3) = (0, 1, 0) = (T(\mathbf{e}_3))_B \text{ therefore } [T]_B = \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

$$\text{Since } (\mathbf{v}_1)_B = (-2, 1, 0), (\mathbf{v}_2)_B = (-1, 0, 1), \text{ and } (\mathbf{v}_3)_B = (0, 1, 0), \text{ we have } P_{B' \rightarrow B} = \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Using Theorem 8.5.2, we obtain

$$[T]_{B'} = P_{B' \rightarrow B}^{-1} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

10. Denoting  $\mathbf{e}_1 = (1, 0, 0)$ ,  $\mathbf{e}_2 = (0, 1, 0)$ , and  $\mathbf{e}_3 = (0, 0, 1)$ , we have  $T(\mathbf{e}_1) = (1, 0, 1) = (T(\mathbf{e}_1))_B$ ,

$$T(\mathbf{e}_2) = (2, -1, 0) = (T(\mathbf{e}_2))_B, \text{ and } T(\mathbf{e}_3) = (-1, 0, 7) = (T(\mathbf{e}_3))_B \text{ therefore } [T]_B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & 7 \end{bmatrix}.$$

$$\text{Since } (\mathbf{v}_1)_B = (1, 0, 0), (\mathbf{v}_2)_B = (1, 1, 0), \text{ and } (\mathbf{v}_3)_B = (1, 1, 1), \text{ we have } P_{B' \rightarrow B} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\text{The inverse of } P_{B' \rightarrow B} \text{ is } P_{B' \rightarrow B}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}. \text{ Using Theorem 8.5.2, we obtain}$$

$$[T]_{B'} = P_{B' \rightarrow B}^{-1} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & 7 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 3 \\ -1 & -2 & -9 \\ 1 & 1 & 8 \end{bmatrix}.$$

11. Denoting  $\mathbf{e}_1 = (1, 0)$  and  $\mathbf{e}_2 = (0, 1)$  we have  $T(\mathbf{e}_1) = (\cos 45^\circ, \sin 45^\circ) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = (T(\mathbf{e}_1))_B$  and

$$T(\mathbf{e}_2) = (-\sin 45^\circ, \cos 45^\circ) = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = (T(\mathbf{e}_2))_B \text{ therefore } [T]_B = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

$$\text{Since } (\mathbf{v}_1)_B = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \text{ and } (\mathbf{v}_2)_B = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \text{ we have } P_{B' \rightarrow B} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}. \text{ Using Theorem 8.5.2, we}$$

$$\text{obtain } [T]_{B'} = P_{B' \rightarrow B}^{-1} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

12. Denoting  $\mathbf{e}_1 = (1, 0)$  and  $\mathbf{e}_2 = (0, 1)$  we have  $T(\mathbf{e}_1) = (1, 0) = (T(\mathbf{e}_1))_B$  and

$$T(\mathbf{e}_2) = (k, 1) = (T(\mathbf{e}_2))_B \text{ therefore } [T]_B = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}.$$

Since  $(\mathbf{v}_1)_B = (k, 1)$  and  $(\mathbf{v}_2)_B = (1, 0)$ , we have  $P_{B' \rightarrow B} = \begin{bmatrix} k & 1 \\ 1 & 0 \end{bmatrix}$ .

The inverse of  $P_{B' \rightarrow B}$  is  $P_{B' \rightarrow B}^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -k \end{bmatrix}$ . Using Theorem 8.5.2, we obtain

$$[T]_{B'} = P_{B' \rightarrow B}^{-1} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} 0 & 1 \\ 1 & -k \end{bmatrix} \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} k & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}.$$

13. Denoting  $\mathbf{p}_1 = 1$  and  $\mathbf{p}_2 = x$  we have  $T(\mathbf{p}_1) = -1 + x$  and  $T(\mathbf{p}_2) = x$ . Thus  $(T(\mathbf{p}_1))_B = (-1, 1)$  and

$$(T(\mathbf{p}_2))_B = (0, 1) \text{ so } [T]_B = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}. \text{ Since } (\mathbf{q}_1)_B = (1, 1) \text{ and } (\mathbf{q}_2)_B = (-1, 1), \text{ we have } P_{B' \rightarrow B} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

The inverse of  $P_{B' \rightarrow B}$  is  $P_{B' \rightarrow B}^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$ . Using Theorem 8.5.2, we obtain

$$[T]_{B'} = P_{B' \rightarrow B}^{-1} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}.$$

14. We have  $T(\mathbf{p}_1) = 9 + 3x$  and  $T(\mathbf{p}_2) = 12 + 2x$ . Thus  $(T(\mathbf{p}_1))_B = (\frac{2}{3}, \frac{1}{2})$  and  $(T(\mathbf{p}_2))_B = (-\frac{2}{9}, \frac{4}{3})$  so

$$[T]_B = \begin{bmatrix} \frac{2}{3} & -\frac{2}{9} \\ \frac{1}{2} & \frac{4}{3} \end{bmatrix}. \text{ Since } (\mathbf{q}_1)_B = (-\frac{2}{9}, \frac{1}{3}) \text{ and } (\mathbf{q}_2)_B = (\frac{7}{9}, -\frac{1}{6}), \text{ we have } P_{B' \rightarrow B} = \begin{bmatrix} -\frac{2}{9} & \frac{7}{9} \\ \frac{1}{3} & -\frac{1}{6} \end{bmatrix}. \text{ The inverse of}$$

$P_{B' \rightarrow B}$  is  $P_{B' \rightarrow B}^{-1} = \begin{bmatrix} \frac{3}{4} & \frac{7}{2} \\ \frac{3}{2} & 1 \end{bmatrix}$ . Using Theorem 8.5.2, we obtain

$$[T]_{B'} = P_{B' \rightarrow B}^{-1} [T]_B P_{B' \rightarrow B} = \begin{bmatrix} \frac{3}{4} & \frac{7}{2} \\ \frac{3}{2} & 1 \end{bmatrix} \begin{bmatrix} \frac{2}{3} & -\frac{2}{9} \\ \frac{1}{2} & \frac{4}{3} \end{bmatrix} \begin{bmatrix} -\frac{2}{9} & \frac{7}{9} \\ \frac{1}{3} & -\frac{1}{6} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

15. (a) We have  $T(1) = 5 + x^2$ ,  $T(x) = 6 - x$ , and  $T(x^2) = 2 - 8x - 2x^2$  so the matrix for  $T$  relative to the

standard basis  $B$  is  $[T]_B = \begin{bmatrix} 5 & 6 & 2 \\ 0 & -1 & -8 \\ 1 & 0 & -2 \end{bmatrix}$ . The characteristic polynomial for  $[T]_B$  is

$$\lambda^3 - 2\lambda^2 - 15\lambda + 36 = (\lambda + 4)(\lambda - 3)^2 \text{ so the eigenvalues of } T \text{ are } \lambda = -4 \text{ and } \lambda = 3.$$

- (b) A basis for the eigenspace of  $[T]_B$  corresponding to  $\lambda = -4$  is  $\begin{bmatrix} -2 \\ \frac{8}{3} \\ 1 \end{bmatrix}$ , so the polynomial

$-2 + \frac{8}{3}x + x^2$  is a basis in  $P^2$  for the corresponding eigenspace of  $T$ .

A basis for the eigenspace of  $[T]_B$  corresponding to  $\lambda = 3$  is  $\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$ , so the polynomial  $5 - 2x + x^2$  is a

basis in  $P^2$  for the corresponding eigenspace of  $T$ .

- 16. (a)** Let  $B$  be the standard basis for  $M_{22}$ ,  $B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ . We have

$$T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, T\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = \begin{bmatrix} 2 & 1 \\ -2 & 0 \end{bmatrix}, T\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \text{ so by}$$

inspection

$$\left[ T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) \right]_B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \left[ T\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) \right]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \left[ T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) \right]_B = \begin{bmatrix} 2 \\ 1 \\ -2 \\ 0 \end{bmatrix}, \left[ T\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) \right]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{therefore } [T]_B = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = A.$$

The characteristic polynomial of  $A$  is

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & 0 & -2 & 0 \\ -1 & \lambda & -1 & 0 \\ 0 & -1 & \lambda + 2 & 0 \\ 0 & 0 & 0 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 (\lambda + 1)(\lambda + 2).$$

We conclude that the eigenvalues of  $T$  are 1, -1, and -2.

- (b)** The reduced row echelon form of  $I - A$  is  $\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$  so that the eigenspace corresponding to

$$\lambda = 1 \text{ contains vectors } \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \text{ where } a = 2s, b = 3s, c = s, d = t. \text{ Vectors } \mathbf{u}_1 = \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix} \text{ and}$$

$\mathbf{u}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$  form a basis for this eigenspace of  $A = [T]_B$ .

The reduced row echelon form of  $-I - A$  is  $\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$  so that the eigenspace corresponding to

$\lambda = -1$  contains vectors  $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$  where  $a = -2t$ ,  $b = t$ ,  $c = t$ ,  $d = 0$ . A vector  $\mathbf{u}_3 = \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \end{bmatrix}$  forms a basis

for this eigenspace of  $A = [T]_B$ .

The reduced row echelon form of  $-2I - A$  is  $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$  so that the eigenspace corresponding to

$\lambda = -2$  contains vectors  $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$  where  $a = -t$ ,  $b = 0$ ,  $c = t$ ,  $d = 0$ . A vector  $\mathbf{u}_4 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$  forms a basis

for this eigenspace of  $A = [T]_B$ .

Reconstructing vectors in  $M_{22}$  from their respective coordinate vectors  $\mathbf{u}_1$ ,  $\mathbf{u}_2$ ,  $\mathbf{u}_3$ , and  $\mathbf{u}_4$  yields

- a basis  $\left\{ \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$  for the eigenspace of  $T$  corresponding to  $\lambda = 1$ ,
- a basis  $\left\{ \begin{bmatrix} -2 & 1 \\ 1 & 0 \end{bmatrix} \right\}$  for the eigenspace of  $T$  corresponding to  $\lambda = -1$ ,
- a basis  $\left\{ \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} \right\}$  for the eigenspace of  $T$  corresponding to  $\lambda = -2$ .

**18.** E.g.,  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  and  $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  are not similar since their determinants do not equal.

19. The matrix of  $T$  with respect to the standard basis for  $R^2$ ,  $B = \{(1,0), (0,1)\}$ , is  $[T]_B = \begin{bmatrix} 3 & -4 \\ -1 & 7 \end{bmatrix}$ .

By Formula (12),  $\det(T) = \det([T]_B) = 17$ .

The characteristic polynomial of  $[T]_B$  is  $\lambda^2 - 10\lambda + 17$ . The eigenvalues of  $T$  are  $5 \pm 2\sqrt{2}$ .

20. The matrix of  $T$  with respect to the standard basis for  $R^3$ ,  $B = \{(1,0,0), (0,1,0), (0,0,1)\}$ , is

$$[T]_B = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}. \text{ By Formula (12), } \det(T) = \det([T]_B) = 0.$$

The characteristic polynomial of  $[T]_B$  is  $\lambda^3 - 3\lambda^2 + 3\lambda$ . The eigenvalues of  $T$  are 0 and  $\frac{3 \pm i\sqrt{3}}{2}$ .

21.  $T(a_0 + a_1x + a_2x^2) = a_0 + a_1(x-1) + a_2(x-1)^2 = (a_0 - a_1 + a_2) + (a_1 - 2a_2)x + a_2x^2$

hence the matrix of  $T$  with respect to the standard basis for  $P_2$ ,  $B = \{1, x, x^2\}$ , is  $[T]_B = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$ .

By Formula (12),  $\det(T) = \det([T]_B) = 1$ . The characteristic polynomial of  $[T]_B$  is  $(\lambda - 1)^3$ .

The only eigenvalue of  $T$  is 1.

22. (a)  $T(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4)$   
 $= a_0 + a_1(2x+1) + a_2(2x+1)^2 + a_3(2x+1)^3 + a_4(2x+1)^4$   
 $= (a_0 + a_1 + a_2 + a_3 + a_4) + (2a_1 + 4a_2 + 6a_3 + 8a_4)x + (4a_2 + 12a_3 + 24a_4)x^2$   
 $+ (8a_3 + 32a_4)x^3 + 16a_4x^4$

hence the matrix of  $T$  with respect to the standard basis for  $P_4$ ,  $B = \{1, x, x^2, x^3, x^4\}$ , is

$$[T]_B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 6 & 8 \\ 0 & 0 & 4 & 12 & 24 \\ 0 & 0 & 0 & 8 & 32 \\ 0 & 0 & 0 & 0 & 16 \end{bmatrix}. \text{ By Theorem 2.1.2, this matrix has nonzero determinant equal to the}$$

product of the entries on the main diagonal. It follows from Theorem 8.2.4 that the rank of  $T$  is 5, and the nullity of  $T$  is 0.

- (b) Theorem 8.2.4 also implies that  $[T]_B$  is invertible, therefore by Theorem 8.4.2,  $T$  is one-to-one.

23. Step (1) follows from the hypothesis (since  $B = P^{-1}AP$ ).

Step (2) follows from Formula (1) in Section 1.4 (since  $I = P^{-1}P$ ).

Step (3) follows from parts (f), (g), and (m) of Theorem 1.4.1.

Step (4) follows from Theorem 2.3.4.

Step (5) follows from commutativity of real number multiplication.

Step (6) follows from Theorem 2.3.4, Formula (1) in Section 1.4, and from Theorem 2.1.2

$$(\text{since } \det(P^{-1})\det(P) = \det(P^{-1}P) = \det(I) = 1).$$

31. Since  $C[\mathbf{x}]_B = D[\mathbf{x}]_B$  for all  $\mathbf{x}$  in  $V$ , then we can let  $\mathbf{x} = \mathbf{v}_i$  for each of the basis vectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$  of  $V$ . Since  $[\mathbf{v}_i]_B = \mathbf{e}_i$  for each  $i$  where  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  is the standard basis for  $R^n$ , this yields  $C\mathbf{e}_i = D\mathbf{e}_i$  for  $i = 1, 2, \dots, n$ . But  $C\mathbf{e}_i$  and  $D\mathbf{e}_i$  are the  $i$ th columns of  $C$  and  $D$ , respectively. Since corresponding columns of  $C$  and  $D$  are all equal,  $C = D$ .

### True-False Exercises

- (a) False. Every matrix is similar to itself since  $A = I^{-1}AI$ .
- (b) True. If  $A = P^{-1}BP$  and  $B = Q^{-1}CQ$ , then  $A = P^{-1}(Q^{-1}CQ)P = (QP)^{-1}C(QP)$ .
- (c) True. Invertibility is a similarity invariant.
- (d) True. If  $A = P^{-1}BP$ , then  $A^{-1} = (P^{-1}BP)^{-1} = P^{-1}B^{-1}P$ .
- (e) True.
- (f) False. For example, if  $T_1$  is the zero operator then  $[T_1]_B$  with respect to any basis  $B$  is a zero matrix.
- (g) True. By Theorem 8.5.2, for any basis  $B'$  for  $R^n$  there exists  $P$  such that  $[T]_{B'} = P^{-1}[T]_B P = P^{-1}IP = P^{-1}P = I$ .
- (h) False. If  $B$  and  $B'$  are different, let  $[T]_B$  be given by the matrix  $P_{B \rightarrow B'}$ . Then  $[T]_{B',B} = P_{B \rightarrow B'}[T]_B = P_{B \rightarrow B'}P_{B' \rightarrow B} = I_n$ .

## 8.6 Geometry of Matrix Operators

1. Coordinates  $(x, y)$  are being transformed to coordinates  $(x', y')$  according to the equation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \text{ Thus } \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} \text{ so } x = x' - 2y' \text{ and } y = -2x' + 5y'.$$

Substituting these into  $y = 4x$  yields  $-2x' + 5y' = 4(x' - 2y')$  or equivalently  $y' = \frac{6}{13}x'$ .

2. Coordinates  $(x, y)$  are being transformed to coordinates  $(x', y')$  according to the equation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \text{ Thus } \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix}^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} \text{ so } x = -2x' + 3y' \text{ and}$$

$y = -3x' + 4y'$ . Substituting these into  $y = -4x + 3$  yields  $-3x' + 4y' = -4(-2x' + 3y') + 3$  or equivalently  $y' = \frac{11}{16}x' + \frac{3}{16}$ .